host or travel?

a study on social connecton among LGBTQ New Yorkers

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Welcome to MPX NYC

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RESPND-MI Collaboration

March 2024, New York City

https://www.mpxnyc.app

Welcome to the MPX NYC Study on the social and sexual connections of LGBTQ New Yorkers. Through this website we report data we collected among LGBTQ New Yorkers in the Fall of 2022 when the first global human mpox oubreak occured.

When we started, MPOX testing was not at all available. Our goal was to estimate the proportion of community members who experienced mpox-like symptoms. We also wanted to understand the structure of social and sexual networks that connect our community so that we could understand how to most effectively distribute MPOX vaccines. Within 5 months of conceiving the project, we had strated collecting data. We kept the survey open for about 1 month, obtaining about 1300 responses.

The data have proven useful - we started sharing periodic analyses with the New York City Department of Health, guiding their outreach activities. Preliminary analyses were shared at the IAS international conference in Montreal, and have through several presentations and public discussions since then.

As MPOX shifted from an outbreak to an epidemic, we took time to come up with the best way to make sense of these data and to convey them to different readers. This report is the result of that effort.

We have prioritized the reproducibility of our study findings and our study design. Our hope is that alliances of LGBTQ activists and researchers will use these methods to answer the pressing questions of their contexts. The data and code we used to make this report are available here [HYPERLINK TO GITHUB]. We have written detailed appendices on the methods we devised to make sense of our data.

The next section gives an overview of what scientists already know about the themes our research report touches on: the relationship between people's social networks and their health, the infectious disease dynamics of a novel outbreak, the health of LGBTQ people, the measurement of spatial data, and the health of LGBTQ people.

1. Housing
2. Nutrition
3. Violence
4. Healthcare access
5. Employment
6. Health

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Part I

Background

1 From outbreak to epidemic

1.1 Rapid oubreak

Mpox cases in the United States (U.S.) were first detected in May 2022 and rapidly escalated, with over 10,000 cases reported by the end of July 2022 (Minhaj et al. 2022). Initial clusters were associated with large gatherings involving sex and concentrated in white, cisgender, gay, bisexual, and other men who have sex with men, with limited data available for minoritized ethno-racial and transgender communities. Over time, new data indicated that Black, Latinx, and transgender communities were also disproportionately affected (Blackburn et al. 2022).

1.2 Slow responses

Despite early warnings from activists about the lack of testing capacity in the U.S., responses from city, state, and federal governments were slow and inadequate (Krellenstein, Osmunson, and Makofane 2023). At the height of Pride month in June, only about 20 mpox tests were administered daily in New York City (NYC), in stark contrast to the 719 probable or confirmed mpox cases in NYC between May 19 to July 15, 2022 (Kyaw et al. 2022). This was a critical period for scaling prevention efforts in vulnerable communities affected by mpox, yet structural divestments led to inadequate data collection; unpredictable vaccination access and vaccine shortages (Anna Grace Lee, n.d.); and challenges in obtaining and providing treatment ("For Monkeypox Patients, Excruciating Symptoms and a Struggle for Care" 2022).

1.3 Weak systems

- Health communication
  - Limited
  - Ugly
  - Stigmatizing
  - Inaccessible
- Healthcare delivery

  - Based on Informal networks // Variable care if you can afford
  - FQHC if you can't afford
- Strategic information
  - Based on biased healthcare delivery data
  - Not geographically specific

1.4 A new pandemic

- Burden
  - Health inequities
  - HIV
- Transmission dynamics
  - Incidence, outbreaks
  - The role of networks

2 Our response

We organized the RESPND-MI study team into three working groups, inviting community members to join regardless of academic credentials: the administration group, the technical group, and the external affairs group. Community partners joined as co-investigators and participated in all collective decision-making.

2.1 Administration

-Study protocol
- IRB - Fundraising - Financial reporting

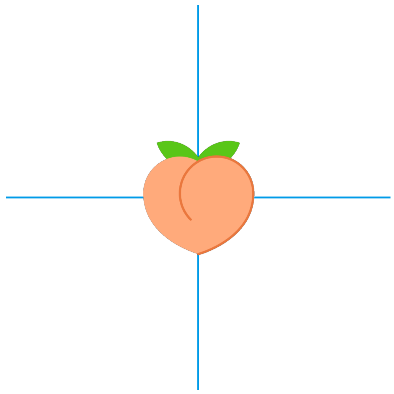
2.2 Technical

- Web development
- Questionnaire development
- Technical expert consultation
- Translation

2.3 External affairs

- Organization outreach
- Marketing development
- Continuous consultation
- Education
- Coordination mechanism

3 Related research



Part II

Methods

4 Data collection

4.1 MPX NYC Person-Place Network Mapper

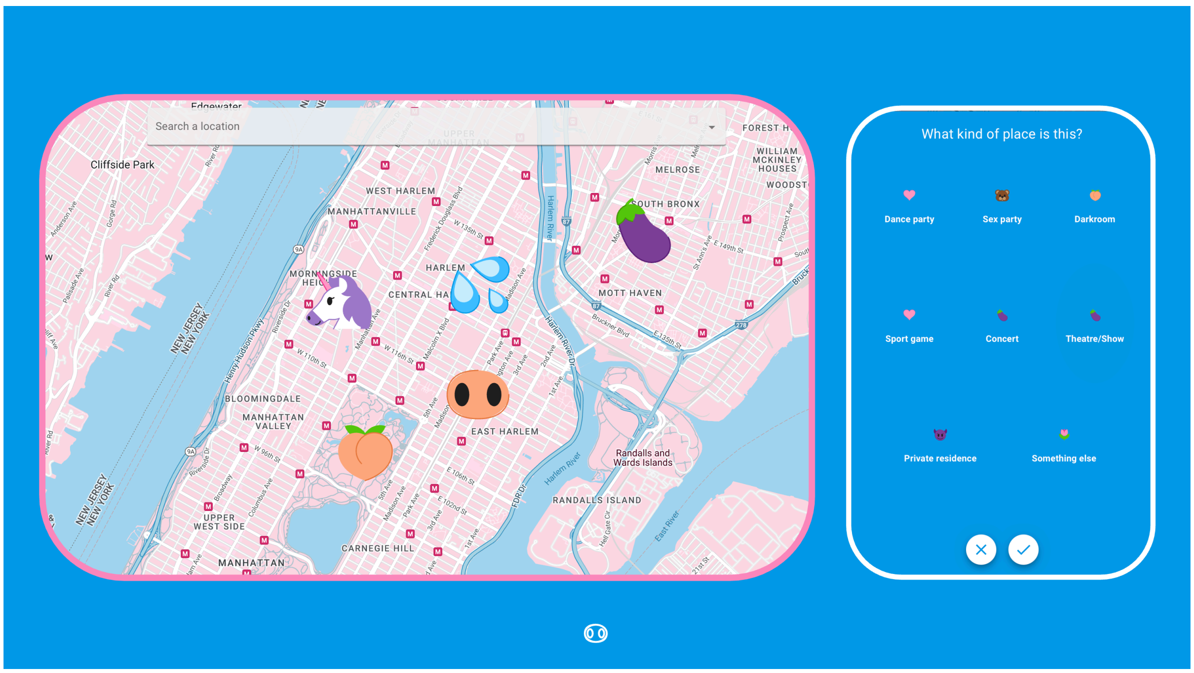


Figure 4.1: Map input question format developed for MPX NYC

Data were collected using a custom designed survey instrument we call the MPX NYC Person-Place Network Mapper. The Person-Place Network Map question was designed to safely elicit sensitive spatial information from survey participants. It displays a map with detailed geographic information including place names, roads, and neighborhood names (see Figure A.1).

To enter spatial information, the participant is prompted to identify places of a certain type by tapping on the map. Each time the participant taps the map, a marker appears where they tapped, and a pop-up window asks questions about the most recently identified location. To ensure anonymity, the survey tool coarsens spatial data to the census tract level before sending them to the study database. Census tracts are non-overlapping spatial units that partition the United States into areas ranging from 2500 to 8000 in population size (US Census 2022).

4.2 MPX NYC marketing and communication

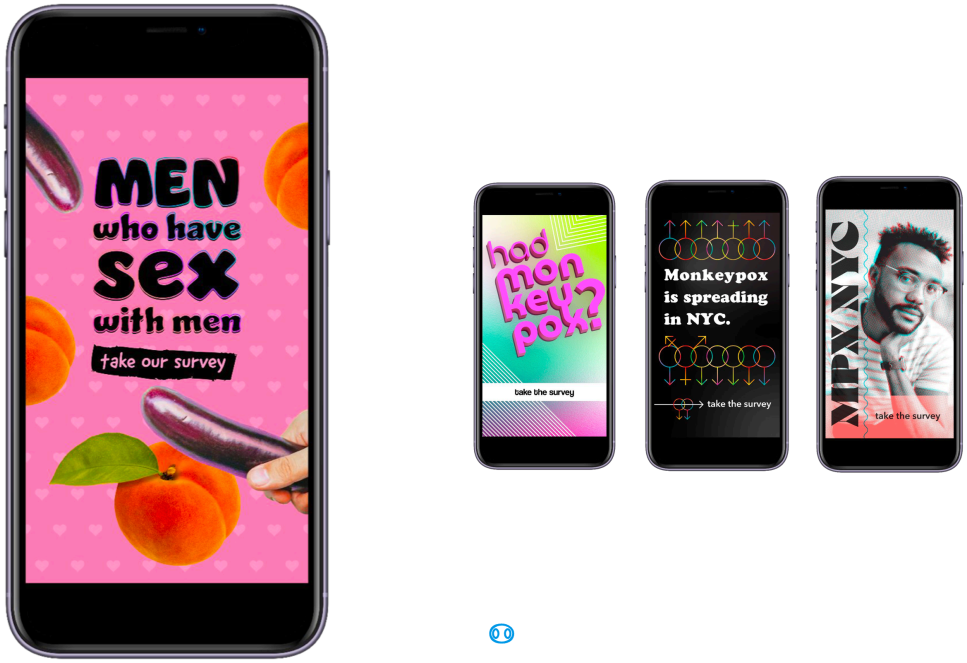


Figure 4.2: Options for visual identity of MPX NYC"

We invested in a bilingual, culturally competent communication campaign in collaboration with a marketing agency. As a starting point, the RESPND-MI team suggested a campaign that playfully uses emojis that have sexual connotations in queer and trans communities, and that integrates LGBT social media influencers as a communication channel. Participants were recruited via paid and donated advertising placed through Grindr, Twitter, Instagram, and outdoor electronic billboards.

4.3 Consent and ethical approval

Prospective participants read and responded to a consent statement before being able to take the survey. The study, including marketing materials and software, was reviewed and approved by the T.H. Chan School of Public Health Institutional Review Board (IRB22-0761).

4.4 Participants and setting

Potential participants were included in the study if they consented to participate, were over 18 years of age, identified as transgender, gay, bisexual, or as men who have sex with other men, and were in New York City in the four weeks preceding their participation. Potential participants were excluded if they were younger than 18 years of age or they withdrew consent.

4.5 Translation

All study materials were professionally translated from English to Spanish. The translation was reviewed by six professionals in health-related fields whose primary language is Spanish.

5 Measures

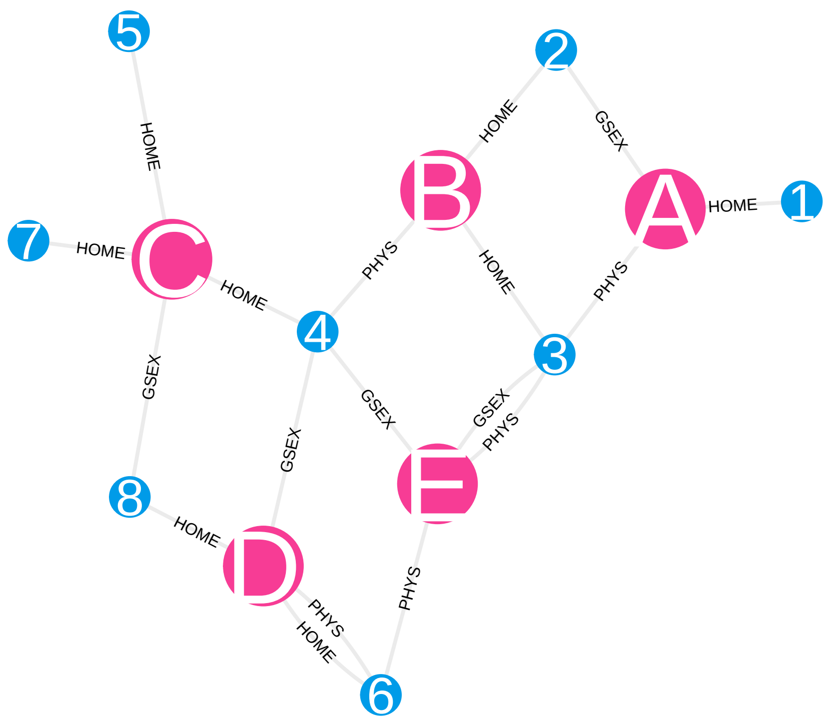


Figure 5.1: Sample person-place network

5.1 Data: MPX NYC participant-community network

The MPX NYC participant community network is a network with community districts and individuals as nodes, and ties between pairs consisting of one individual and one community district. While spatial data were collected using census tract identifiers, the unit of spatial analysis is the community district. Community districts are spatial units that divide New York City into regions which are nested by the five boroughs of New York City, and which in turn nest census tracts.

A tie indicates that the individual as some relation with the community district. This relation can be residence or group contact. Related to this network are the community district network and participant network, described below.

5.1.1 MPX NYC community district network

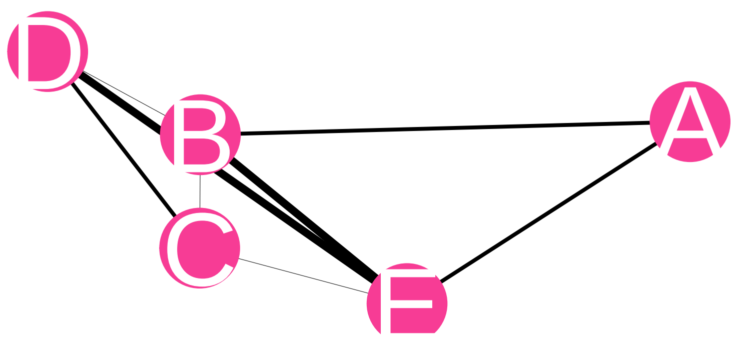


Figure 5.2: Sample place-place network

The MPX NYC community district network is a network with community districts as nodes. A weighted tie between two nodes indicates that the two community districts both have at least one MPX NYX participant who has a residence or community district in both places. The weight is given by the count of such individuals. Of interest is the influence each community district has in city-wide outbreak dynamics. It is measured using community district social and spatial reach.

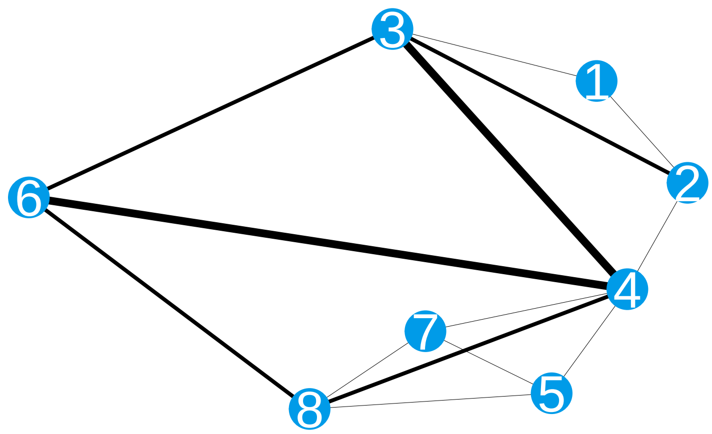


Figure 5.3: Sample person-person network

5.1.2 MPX NYC participant network

We examined the patterning of shared physical context among participants using the MPX NYC participant network. In this network, survey participants are represented as nodes. A weighted edge indicates that the individuals represented by the nodes have at least one residence or contact venue in a common community district. The edge weight is given by the count of community districts the pair share. Of interest are the overall connectedness of the network and the pattern of connection across different subgroups.

5.2 Node characteristics

5.2.1 Individual measures

We measured participant demographics age, race, sex assigned at birth, gender identity, sexual orientation, and community district of residence; health-related questions on HIV status, viral suppression, STI-related symptoms, PrEP use, and mpox vaccination status; and questions related to social and sexual connection with queer and transgender communities: count of recent individual sex contacts, count of recent individual physical (non-sexual) contacts, count of recent important social contacts, and any recent physical or sexual contact in a group setting.

5.2.1.1 Gender modality

Gender modality was measured using a two-step approach: first, we solicited the participant's current gender (man, woman, trans man, trans woman, non-binary, other), and then sex assigned at birth (male, female, other). We then constructed a gender modality variable as follows:

Table 5.1: _construction of gender modality variable

Level	Condition
Cisgender man	gender is 'man' and assigned sex is 'male'
Cisgender woman	gender is 'woman' and assigned sex is 'female'
Transgender man	gender is 'man' and assigned sex is 'female'
Transgender woman	gender is 'woman' and assigned sex is 'male'
Transgender man	gender is 'trans man' and sex is any value
Transgender woman	gender is 'trans woman' and sex is any value
Non-binary	gender is 'non-binary' and sex is any value
Another demographic	gender is 'other' and sex is any value

5.2.1.2 Race-gender

We created a Race-gender variable which is a categorical variable with levels identical to the gender modality with one modification: instead of having all cisgender men in one level we divided the group up into white cisgender men, Latinx cisgender men, Black cisgender men, and under the category Other cisgender men, we grouped Asian, Pacific Islander and other cisgender men.

5.2.2 Community district measures

5.2.2.1 Social reach

For a given community district, social reach is defined as the sum of edge weights for edges connecting that community district to individuals (in the MPX NYC spatial participant network). It quantifies the volume or level of contact in the community district.

5.2.2.2 Spatial reach

For a given community district spatial reach is defined as the sum of edge weights for edges connecting that community district to others (in the MPX NYC community district network. It quantifies the volume or level of exchange of individuals between a particular community district and others.

5.3 Edge characteristics

5.3.1 Residence edges

All participants were asked the location of their residence through the Person-Place Network Map question. Each participant's residences is represented as an edge between the participant and the community district containing their residence.

5.3.2 Contact venue edges

Participants who indicated recent physical or sexual contact in a group setting were asked for the locations of each of the places where the contact happened. We call these contact places. They are each represented as an edge between the participant and the community district containing the contact place.

5.3.2.1 Venue type

Participants were asked the type of venue each contact place was (dance party, sex party, darkroom, sport game, concert, theatre/show, private residence, sex club, bathhouse, park, something else).

5.3.2.2 Sexual contact

Participants were also asked if they had sexual contact at each contact place.

5.3.2.3 Distance from home

We constructed an approximate measure of the distance between residence and contact venue by categorizing venues into those that are in the same community district as the residence; in a different community district but the same borough; or in a different borough. We refer to this measure as distance from home.

5.4 Network characteristics

5.4.0.1 Size of largest connected component

We measure connectedness using the proportion of nodes in the largest connected component (LCC) of the MPX NYC participant network. In a network, a connected component is a subset of nodes such that each node in the subset is reachable from any other node in the subset, and such that there are no nodes outside the subset which are reachable from a node inside it. By "reachable", we mean possible to reach by tracing a path from one node to another over existing network ties. We consider a network with a larger LCC to be better connected than a network with a smaller LCC.

5.4.0.2 Spatial mixing ratio

We measure patterns of mixing between a pair of distict subgroups of participants using the spatial mixing ratio. It is a positive real number which is greater than 1 if individuals in the first subgroup tend to live and play in community districts that individuals in the second subgroup do. It is defined as the average (calculated across individuals) spatial preference of the first group for the second group, divided by the prevalence of the second group. For a given individual, spatial preference for a given group is defined as the sum of the weights of the edges connecting to nodes of the second group, divided by the total sum weights.

6 Statistical Analysis

6.1 Descriptive statistics

We tabulated participant characteristics against group contact, and visualized the distribution of sexual orientation categories across levels of the race-gender variable. We also tabulated contact venue characteristics against sexual contact, and further visualized venue type against distance from home in addition to sexual contact.

To understand how participants move across community districts from home to contact places, we cross-tabulated home community district (the community district of the participants' residence) against contact place community district and visualized the resulting table. To identify community districts that tend to receive more participants from residences in outside community districts, and those which tend to send participants, we subtracted the proportion of residences in each community district from the proportion of contact venues. The result of this calculation is shown as a bar graph.

Finally, we calculated the spatial mixing coefficient across groups defined by age, sexual orientation, and race-gender to understand how individuals in subgroups defined by these variables interact in space.

To illustrate how the pattern of interaction among queer and trans people should inform early responses to novel infectious diseases, we examined the impact of a simplified, spatially-targeted intervention to immunize the population represented by the MPX NYC participants against a novel pathogen. We examine the impact of two different intervention strategies on two population outcome measures of interest

6.2 Outbreak control strategies

In this hypothetical intervention, community districts are ordered by priority and are immunized one at a time, beginning with the highest priority community district. We assume that to immunize a community district is to immunize each individual who has a contact venue or residence in that community district. We assume that once a person is immunized, they are unable to acquire or transmit the pathogen, and we assume that everyone can only be immunized once. We assessed two strategies for prioritizing community districts: the contact-neutralizing strategy and the movement-neutralizing strategy.

6.2.1 Contact neutralizing strategy

In the contact-neutralizing strategy, we rank districts according to the number of contact venues and residences they contain (counting only the residences of participants who reported group contact, and who are not yet immunized). This strategy prioritizes community districts with a high amount of sexual or physical contact for immunization. It attempts to immunize as many people in as few community districts as possible.

6.2.2 Movement neutralizing strategy

In the movement-neutralizing strategy, we rank community districts by their spatial centrality. This strategy prioritizes community districts with a high level of connection to other community districts for immunization. It attempts to immunize the community districts that are potential hubs for the dissemination of the pathogen across the city early to geographically contain the outbreak.

6.3 Outbreak control outcomes

For each community district immunized, we updated the MPX NYC participant network by removing nodes that represent immunized individuals. We then calculated the two outcomes each time: spatial efficiency, and spatial network impact.

6.3.1 Spatial efficiency

Spatial efficiency is defined as the total count of immunized individuals divided by the count of immunized community districts. To illustrate differences in spatial efficiency across intervention strategies, we visualized the number of community districts it would take to immunize 33% of the population, and 66% of the population respectively.

6.3.2 Spatial network impact

Spatial network impact is defined as the proportion of individuals included in the largest connected component of the MPX NYC participant network. To illustrate differences in spatial network impact across intervention strategies, for each additional community district immunized, we plotted the proportions of individuals who are immunized, unimmunized and in the largest connected component, and unimmunized but outside of the largest connected component, respectively.

6.4 Statistical inference

We used the bootstrap to estimate confidence intervals, resampling participants 5 times with replacement.

Part III

Results

7 People

The MPX NYC survey was open from 30 August to 15 November, 2022, but recruited 70% participants in two 48 hour periods: half were recruited from 30 to 31 Aug, and another 20% from 10 to 11 September (see **?@fig-people-descriptives**). Over these periods, the MPX NYC recruitment advertisements were shown to users of Grindr in New York City upon logging on to the app.

MPX NYC participants are demographically diverse. Over 50% of participants were between the ages of 18-34 at the time of the survey. Two fifths were white cisgender men, about 15% were latinx cisgender men, and 10% were black cisgender men. About 5% of participants were transgender (with slightly over half being transgender men). Non-binary individuals comprised 7% of participants. Two-thirds of participants identified as gay, 14% as bisexual, and 12% as queer. Just under half of participants resided in the borough of Manhattan, and one third in Brooklyn.

In terms of health status and health services, one in ten participants reported living with HIV - the overwhelming majority were virally suppressed. In addition, about 16% reported at least one STI symptom. Two out of five participants were on PrEP at the time they took the survey. Two-thirds had received the MPOX vaccine, though only 8% had received MPOX testing.

Table 7.1: Participants in MPX NYC Study by Sex / Physical Contact in Congregate Setting in past 4 weeks

	Physical or sexual contact in group setting in past 4 weeks		
Variable N = 530[1] N = 774[1] N = 1,304[1]	**Yes** **No** **Overall** **p-value**[2]		
Age			
<18	0 (0%)	0 (0%)	0 (0%)
18-24	92 (17%)	103 (13%)	195 (15%)
25-34	234 (44%)	304 (39%)	538 (41%)
35-44	135 (25%)	190 (25%)	325 (25%)
45-54	48 (9.1%)	97 (13%)	145 (11%)
55+	21 (4.0%)	80 (10%)	101 (7.7%)
Race			

White	279 (53%)	416 (54%)	695 (53%)
Latinx	103 (19%)	128 (17%)	231 (18%)
Black	60 (11%)	96 (12%)	156 (12%)
Asian	32 (6.0%)	52 (6.7%)	84 (6.5%)
Multi Racial	37 (7.0%)	42 (5.4%)	79 (6.1%)
Another Group	19 (3.6%)	38 (4.9%)	57 (4.4%)
Unknown	0	2	2
Gender identity			
Cisgender Man	427 (81%)	630 (81%)	1,057 (81%)
Non Binary	39 (7.4%)	57 (7.4%)	96 (7.4%)
Transgender Man	17 (3.2%)	17 (2.2%)	34 (2.6%)
Transgender Woman	14 (2.6%)	15 (1.9%)	29 (2.2%)
Cisgender Woman	19 (3.6%)	43 (5.6%)	62 (4.8%)
Another Demographic	14 (2.6%)	12 (1.6%)	26 (2.0%)
Race-gender			
White cisgender man	218 (41%)	334 (43%)	552 (42%)
Latinx cisgender man	88 (17%)	110 (14%)	198 (15%)
Black cisgender man	50 (9.4%)	81 (10%)	131 (10%)
Transgender man	17 (3.2%)	17 (2.2%)	34 (2.6%)
Transgender woman	14 (2.6%)	15 (1.9%)	29 (2.2%)
Non binary	39 (7.4%)	57 (7.4%)	96 (7.4%)
Cisgender woman	19 (3.6%)	43 (5.6%)	62 (4.8%)
Other cisgender man	71 (13%)	105 (14%)	176 (13%)
Another demographic	14 (2.6%)	12 (1.6%)	26 (2.0%)
Sexual orientation			
Gay	362 (68%)	498 (64%)	860 (66%)
Bisexual	67 (13%)	119 (15%)	186 (14%)
Straight	14 (2.6%)	50 (6.5%)	64 (4.9%)
Queer	71 (13%)	88 (11%)	159 (12%)
Something Else	16 (3.0%)	19 (2.5%)	35 (2.7%)
Home borough			
Bronx	31 (5.8%)	66 (8.5%)	97 (7.4%)
Brooklyn	176 (33%)	258 (33%)	434 (33%)
Manhattan	259 (49%)	320 (41%)	579 (44%)
Queens	58 (11%)	119 (15%)	177 (14%)
Staten Island	6 (1.1%)	11 (1.4%)	17 (1.3%)
Recruitment channel			
Grindr	262 (49%)	444 (57%)	706 (54%)
Partner toolkit	60 (11%)	82 (11%)	142 (11%)
Instagram	47 (8.9%)	65 (8.4%)	112 (8.6%)
Twitter	19 (3.6%)	43 (5.6%)	62 (4.8%)
Unknown	142 (27%)	140 (18%)	282 (22%)
Recent important social contacts			

0	64 (12%)	155 (20%)	219 (17%)
1-5	206 (39%)	333 (43%)	539 (41%)
6-10	136 (26%)	167 (22%)	303 (23%)
11-15	49 (9.2%)	41 (5.3%)	90 (6.9%)
16+	75 (14%)	78 (10%)	153 (12%)
Recent physical contacts			
0	98 (18%)	301 (39%)	399 (31%)
1	68 (13%)	191 (25%)	259 (20%)
2	66 (12%)	105 (14%)	171 (13%)
3	46 (8.7%)	64 (8.3%)	110 (8.4%)
4	47 (8.9%)	32 (4.1%)	79 (6.1%)
5+	205 (39%)	81 (10%)	286 (22%)
HIV status			
Living with HIV	60 (11%)	81 (10%)	141 (11%)
Not living with HIV	447 (84%)	676 (87%)	1,123 (86%)
Unsure	23 (4.3%)	17 (2.2%)	40 (3.1%)
HIV supressed			
Yes	54 (90%)	78 (95%)	132 (93%)
No	3 (5.0%)	4 (4.9%)	7 (4.9%)
Unsure	3 (5.0%)	0 (0%)	3 (2.1%)
Unknown	470	692	1,162
Number of symptoms (calculated)			
0	412 (78%)	681 (88%)	1,093 (84%)
1+	118 (22%)	93 (12%)	211 (16%)
PrEP user	252 (54%)	253 (37%)	505 (43%)
Unknown	60	81	141
Mpox testing	58 (11%)	47 (6.1%)	105 (8.1%)
Recent individual sex contacts			
0	48 (9.1%)	272 (35%)	320 (25%)
1	86 (16%)	258 (33%)	344 (26%)
2	96 (18%)	110 (14%)	206 (16%)
3	82 (15%)	62 (8.0%)	144 (11%)
4	62 (12%)	29 (3.7%)	91 (7.0%)
5+	156 (29%)	43 (5.6%)	199 (15%)
Hookup travel time			
0-15 min	121 (23%)	262 (34%)	383 (29%)
16-30 min	221 (42%)	270 (35%)	491 (38%)
31-45 min	71 (13%)	88 (11%)	159 (12%)
45+ min	117 (22%)	154 (20%)	271 (21%)
Survey date			
30-31 Aug 2022	239 (45%)	372 (48%)	611 (47%)
01-10 Sep 2022	110 (21%)	118 (15%)	228 (17%)
10-11 Sep 2022	98 (18%)	171 (22%)	269 (21%)

12 Sep - 31 Nov 2022	83 (16%)	113 (15%)	196 (15%)
Mpox vaccine			
Yes	409 (77%)	461 (60%)	870 (67%)
No	116 (22%)	311 (40%)	427 (33%)
Unsure	5 (0.9%)	2 (0.3%)	7 (0.5%)

[1]n (%)
[2]Pearson's Chi-squared test; Fisher's exact test
Data source: MPX NYC 2022

This diversity of subgroups implies different strategies for communication, and possibly different health interventions. For instance, the sexual orientation of participants differed markedly by gender (see Figure 7.1). The vast majority of cisgender men identified as gay, with some identifying as bisexual. Transgender men, on the other hand, were most likely to identify as queer, followed by bisexual or gay. Transgender women tended to identify as straight, bisexual, or other sexual orientations. Cisgender women participants tended to be straight, followed by bisexual, followed by queer. For non-binary people, most identified as queer, followed by gay and bisexual.

MPX NYC participants were deeply connected to queer and trans communities in New York City. About two-thirds reported being in recent (preceding 4 weeks) contact with 1 - 10 queer or trans friends. Only 20% reported not being in contact. About half of the participants had between 1 and 3 recent serial sex partnerships with one fifth reporting four or more partners. About one third of participants reported spending 16-30 minutes when traveling to a hookup while a third spent more than 30 minutes. Almost one in four participants reported having recent physical (non-sexual) contact with 2 or more individuals at the same time, and one third reported having had no contact. [DOUBLE CHECK]

Among all MPX NYC participants, 40% (530) reported reported having had sex or physical contact with two or more people at the same time. These participants identified 604 venues at which they had contact. These contact venues are examined in the next section.

Participants who reported recent physical or sexual contact in a group setting (group contact) were slightly younger than those who did not. They were more likely to identify as gay and less likely to identify as straight. Surprisingly, they were less likely to have been recruited through a Grindr advertisement and more likely to have been recruited before the first Grindr pop-up advertisement on 30 August 2022.

These participants did not differ from the rest in terms of HIV status or HIV viral suppression, but they were more likely to have at least one STI symptom, and showed higher utilization of PrEP, mpox testing, and mpox vaccine. They were more closely connected with the queer and trans community, reporting more contact with friends, more physical contact partners and more sexual contact partners. They spent more time travelling to hookups than participants who had not had recent contact in a group setting.

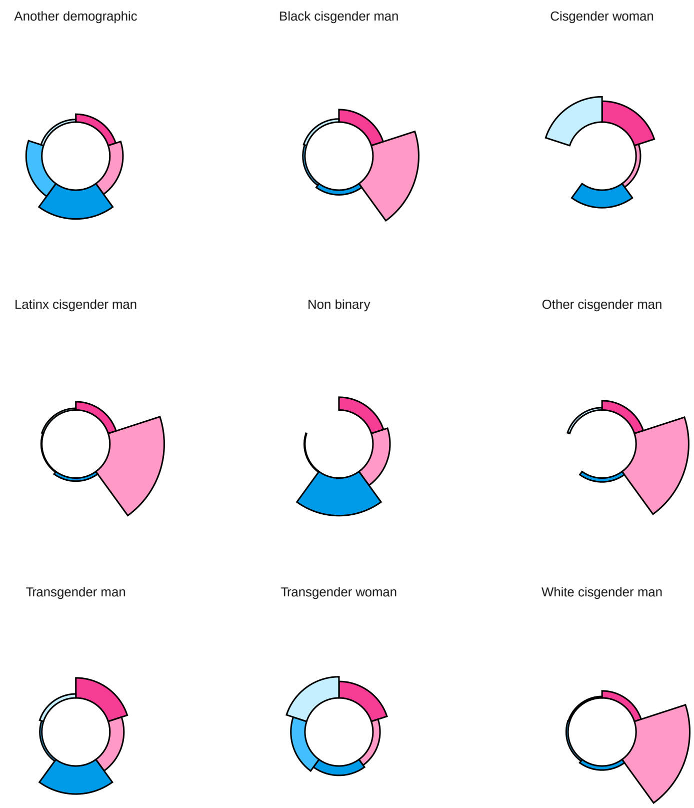
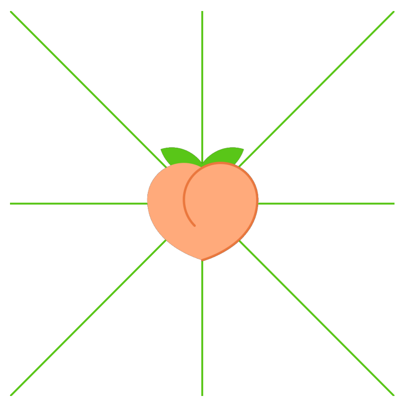


Figure 7.1: Sexual orientation of MPX NYC participants by race and gender



8 Places

Of the 604 locations at which participant had physical or sexual contact in a group setting (contact venues), about two out of five were private residences, and about a quarter were dance parties. More than half of the contact venues were in Manhattan, while less than a third were in the Brooklyn. A quarter of all contact venues were in the same community as the participant's residence, a third were in a different borough, and the remainder were in the same borough as home, but a different community district.

Table 8.1: Contact venue by venue and contact type

	Sexual contact at contact venue			
Variable N = 264[1] N = 352[1] N = 616[1]	**Did not have sex** **Had sex** **Overall** **p-value**[2]			
Place type				<0.001
Concert/Theatre/Show	63 (24%)	4 (1.1%)	67 (11%)	
Dance Party	110 (42%)	33 (9.4%)	143 (23%)	
Dark Room/Sex Party	10 (3.8%)	52 (15%)	62 (10%)	
Private Residence	28 (11%)	222 (63%)	250 (41%)	
Something Else	40 (15%)	40 (11%)	80 (13%)	
Sport Game	13 (4.9%)	1 (0.3%)	14 (2.3%)	
Place borough				0.3
Bronx	6 (2.3%)	18 (5.1%)	24 (3.9%)	
Brooklyn	84 (32%)	96 (27%)	180 (29%)	
Manhattan	144 (55%)	188 (53%)	332 (54%)	
Queens	29 (11%)	48 (14%)	77 (13%)	
Staten Island	1 (0.4%)	2 (0.6%)	3 (0.5%)	
Distance from home				<0.001
Same Community District	42 (16%)	144 (41%)	186 (30%)	
Same Borough	116 (44%)	114 (32%)	230 (37%)	
Different Borough	106 (40%)	94 (27%)	200 (32%)	

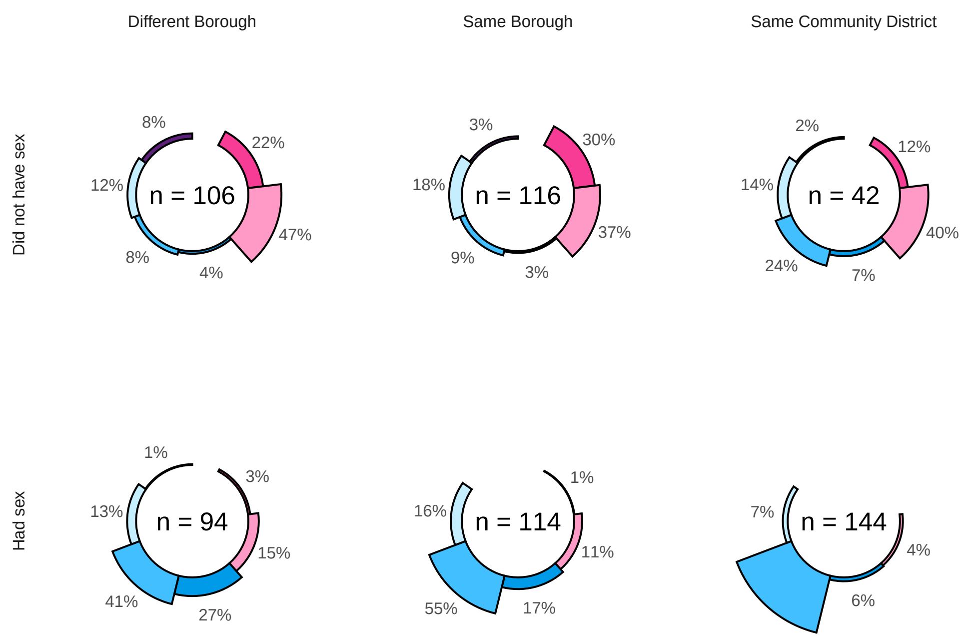
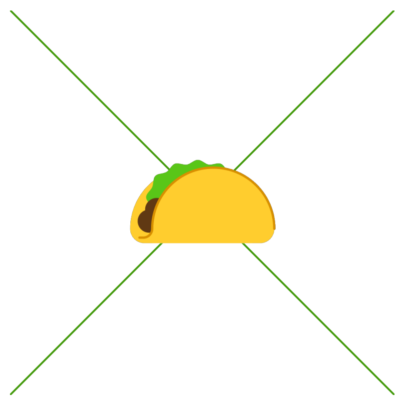


Figure 8.1: MPX NYC contact venue type by distance from home and sexual contact

[1] n (%)
[2] Pearson's Chi-squared test; Fisher's exact test
Data source: MPX NYC 2022

Contact venues at which participants had sexual contact (rather than non-sexual physical contact) were overwhelmingly private residences whereas contact venues at which they had non-sexual physical contact were mostly dance parties (see Figure 8.1). The closer to home a location was, the more likely it was to be a private residence and the more likely it was the site of sexual contact. By contrast, the further from home a contact venue was, the more likely it was to be a dance party.



9 Movement

9.1 Movement between home to contact venues

9.1.1 Spatial patterns

Figure 9.2 shows the difference betweeen the proportion of primary residences in each community district and the proprotion of contact venues in each community district. It shows that within Brooklyn, Manhattan, and Queens, there is a small set of community districts which tend to be markedly over-represented among contact venues compared to their share of primary residences. These are BK01, MN05, and QN05, respectively.

9.1.2 Spatial concentration

Figure 9.1 maps each contact venue's community district (x-axis) against the participant's home community district (y-axis). The size of the dot is proportional to the total count of contact venues with the given home and contact venue community district. The color of the dot represents the borough of the contact venue.

The dots on the diagonal line from top left to bottom right, show prominently, indicating that a sizeable proportion of movement happens within community districts. The figure also shows four prominent clusters - the cluster at the top left (in pink) shows significant movement between home and contact venues within Brooklyn and the cluster towards the middle of the plot (in purple) shows significant movement within Manhattan. The clusters at the top center and bottom left show significant movement from Brooklyn to Manhattan and from Manhattan to Brooklyn, respectively.

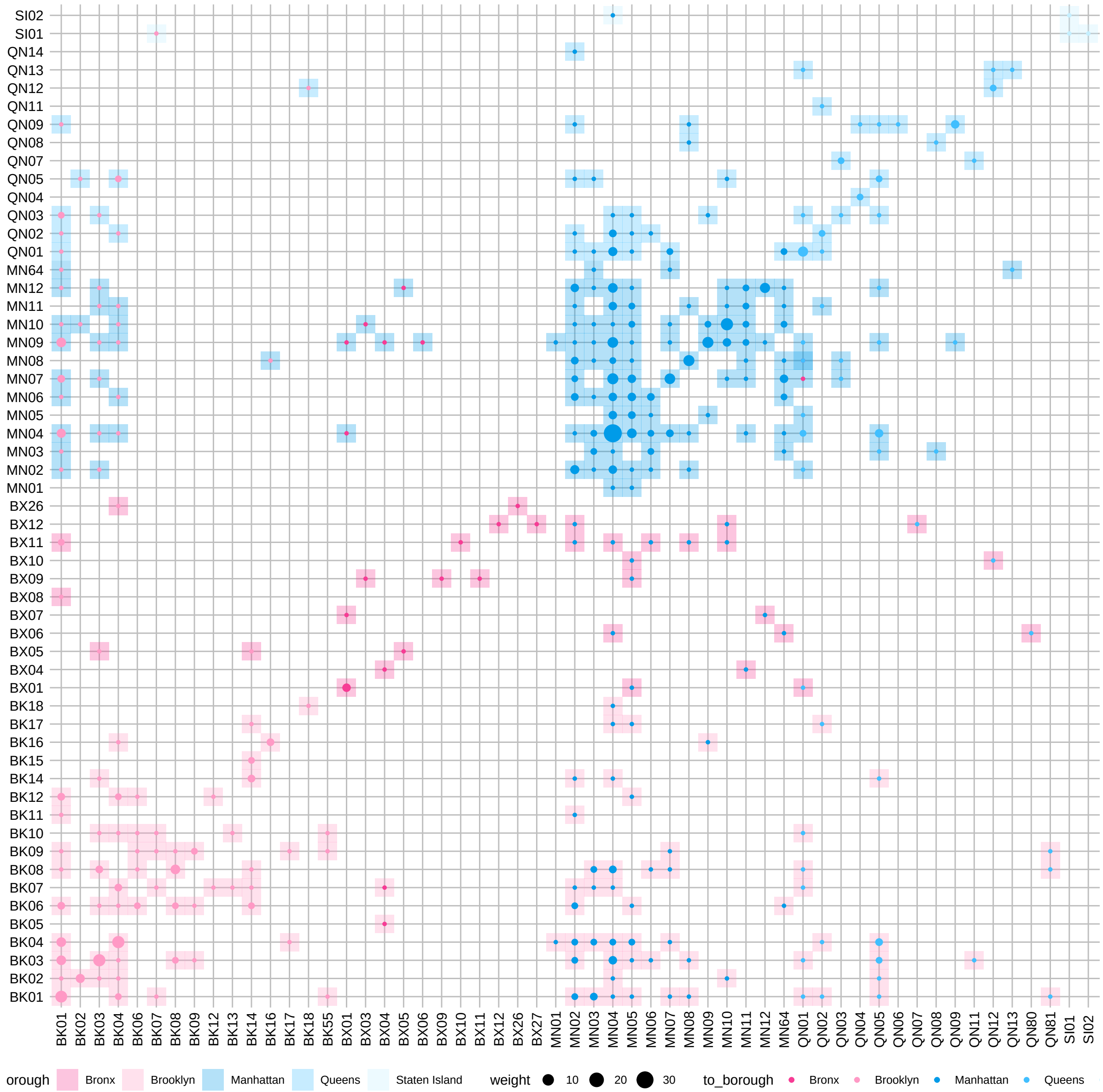


Figure 9.1: Movement between home and contact venues by MPX NYC participant. The figure maps each contact venue's community district (x-axis) against the participant's home community district (y-axis). The size of the dot is proportional to the total count of contact venues with the given home and contact venue community district. The color of the dot represents the borough of the contact venue.

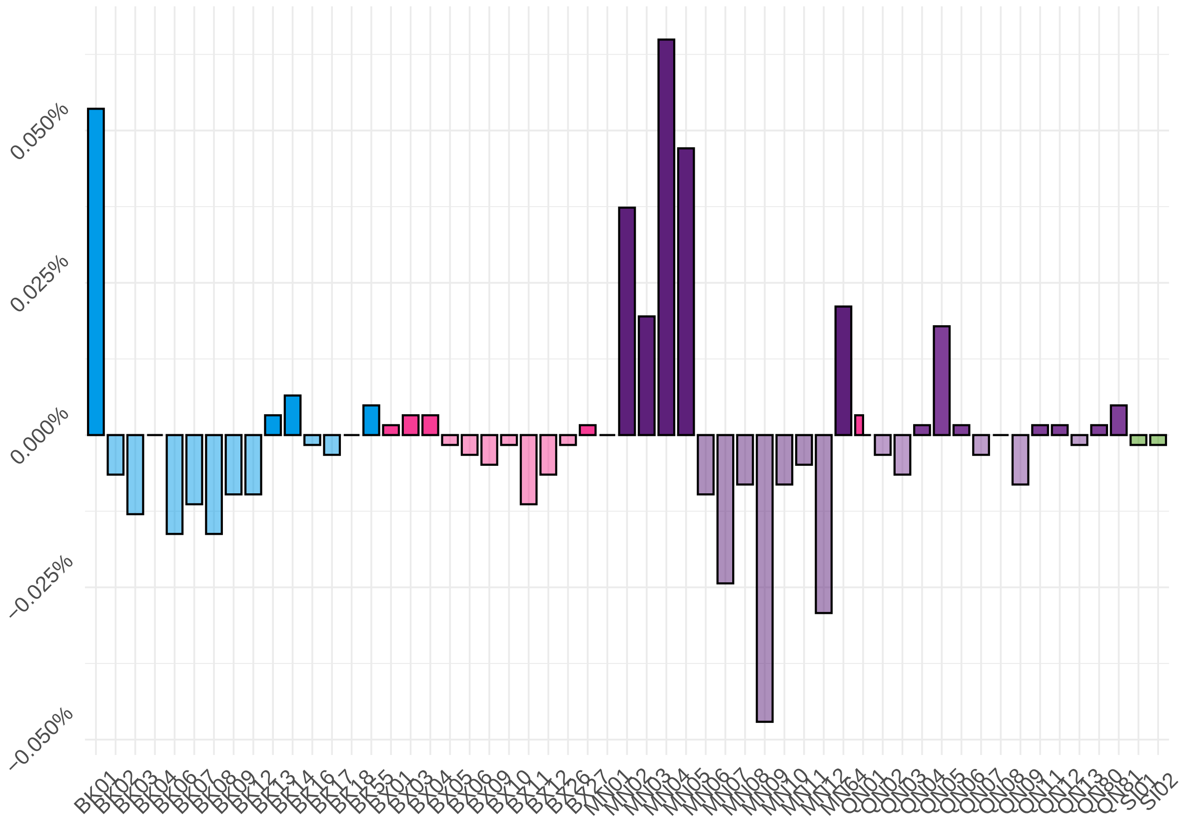
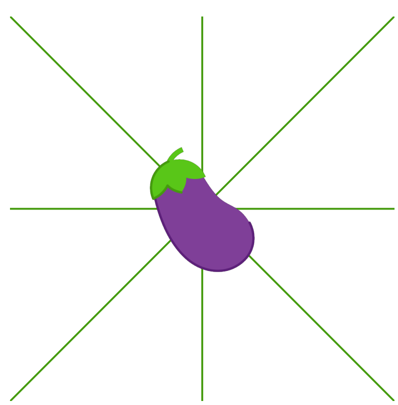


Figure 9.2: Overall Movement between home and contact venues by MPX NYC participant. The figure shows each community districts over-representation among contact venues compared to residences



10 Mixing

Race-gender

?@fig-subgroup-mixing quantifies the degree to which individuals with characteristics shown on the left live or play in an overlapping set of community districts with individuals whose characteristics are shown at the top. Overall, MPX NYC participants showed spatial homophily - the spatial selection coefficient from a given group to the same group was positive for all levels of race-gender, sexual orientation, and age. In addition, black cisgender men and non-binary persons tended to visit or live in different neighborhoods than white cisgender men. Non-binary persons also showed a preference for community districts in which transgender men live. Gay identified participants preferred to live and play in community districts where bisexual, straight and queer identified participants did not live. Similarly, those groups selected against community districts in which gay identified individuals live or play.

Sexual orientation

Age

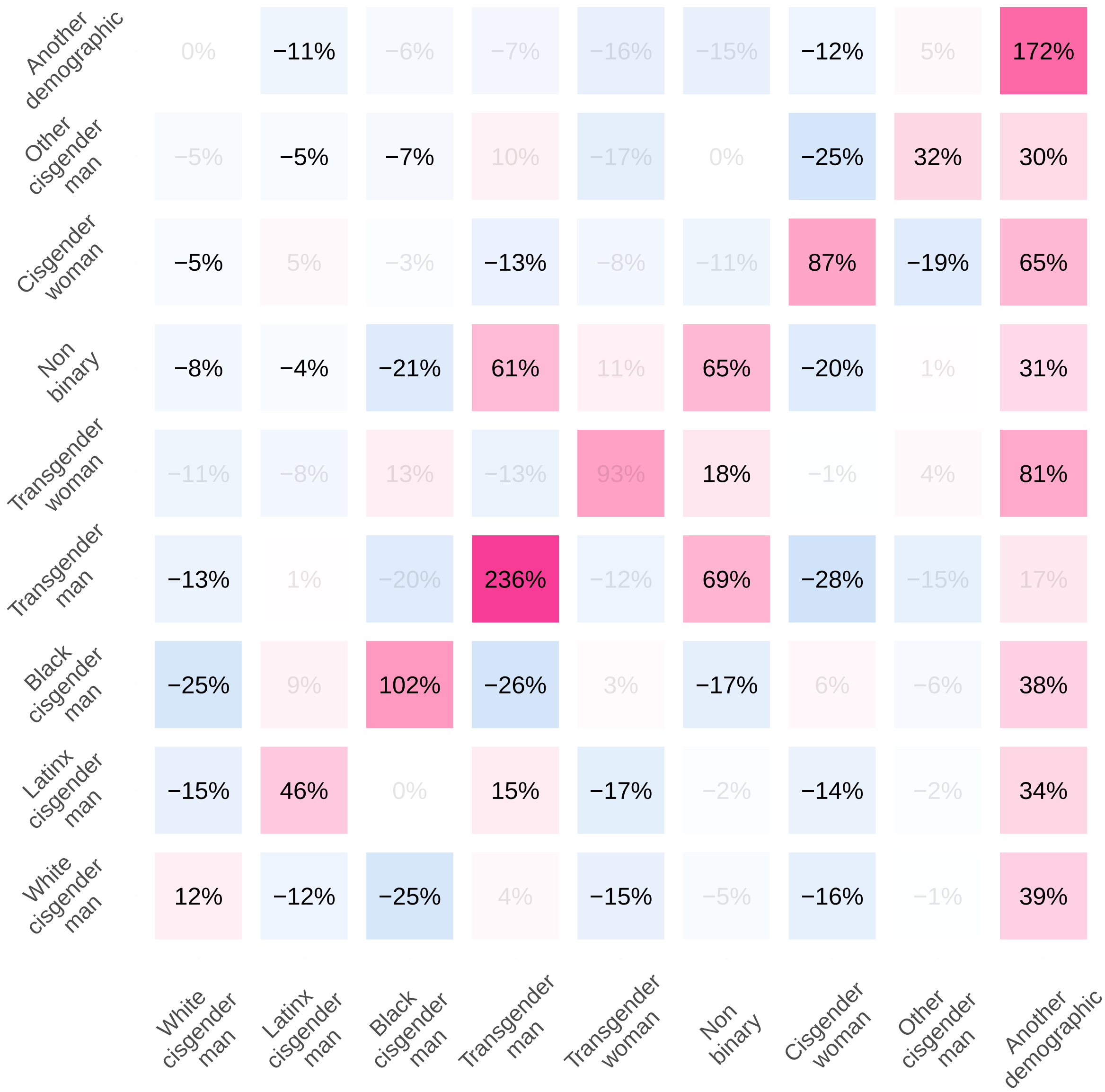


Figure 10.1: Social mixing matrices for MPX NYC participants by race-gende

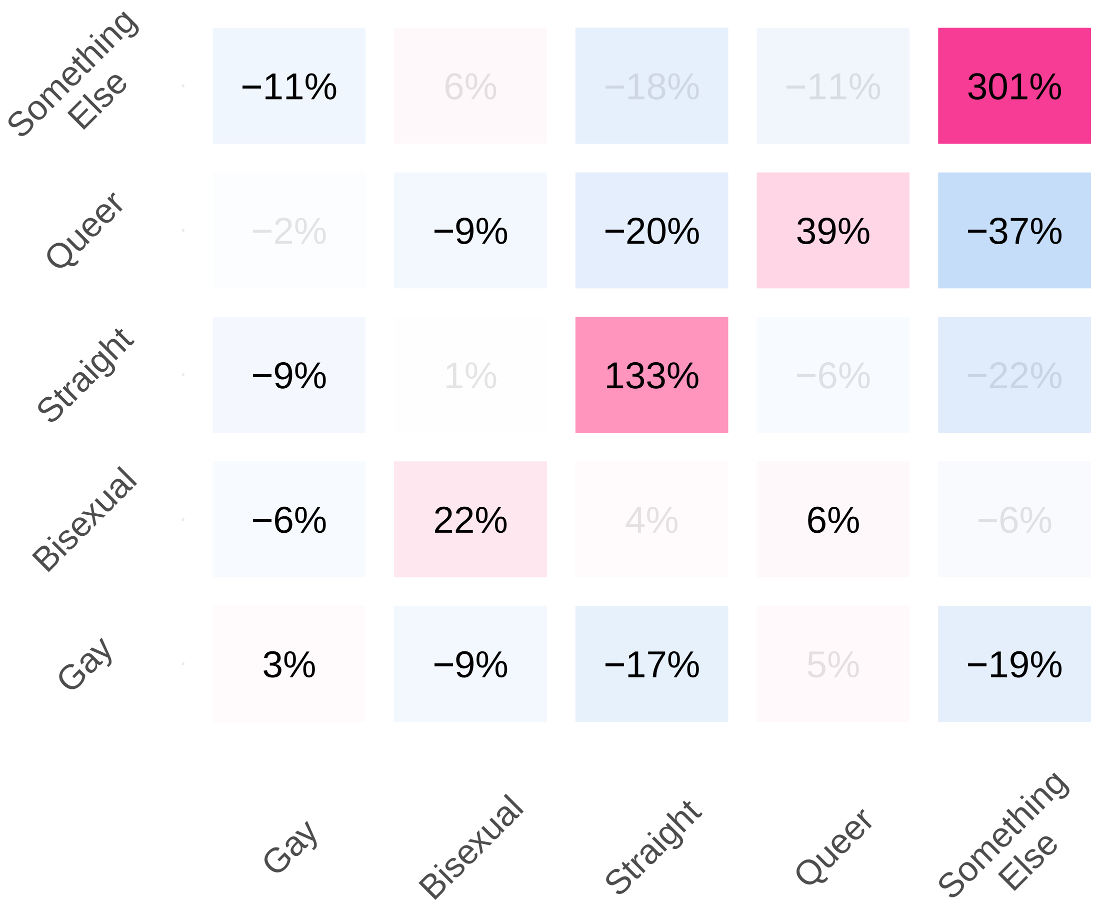


Figure 10.2: Social mixing matrix for MPX NYC participants by sexual orientation

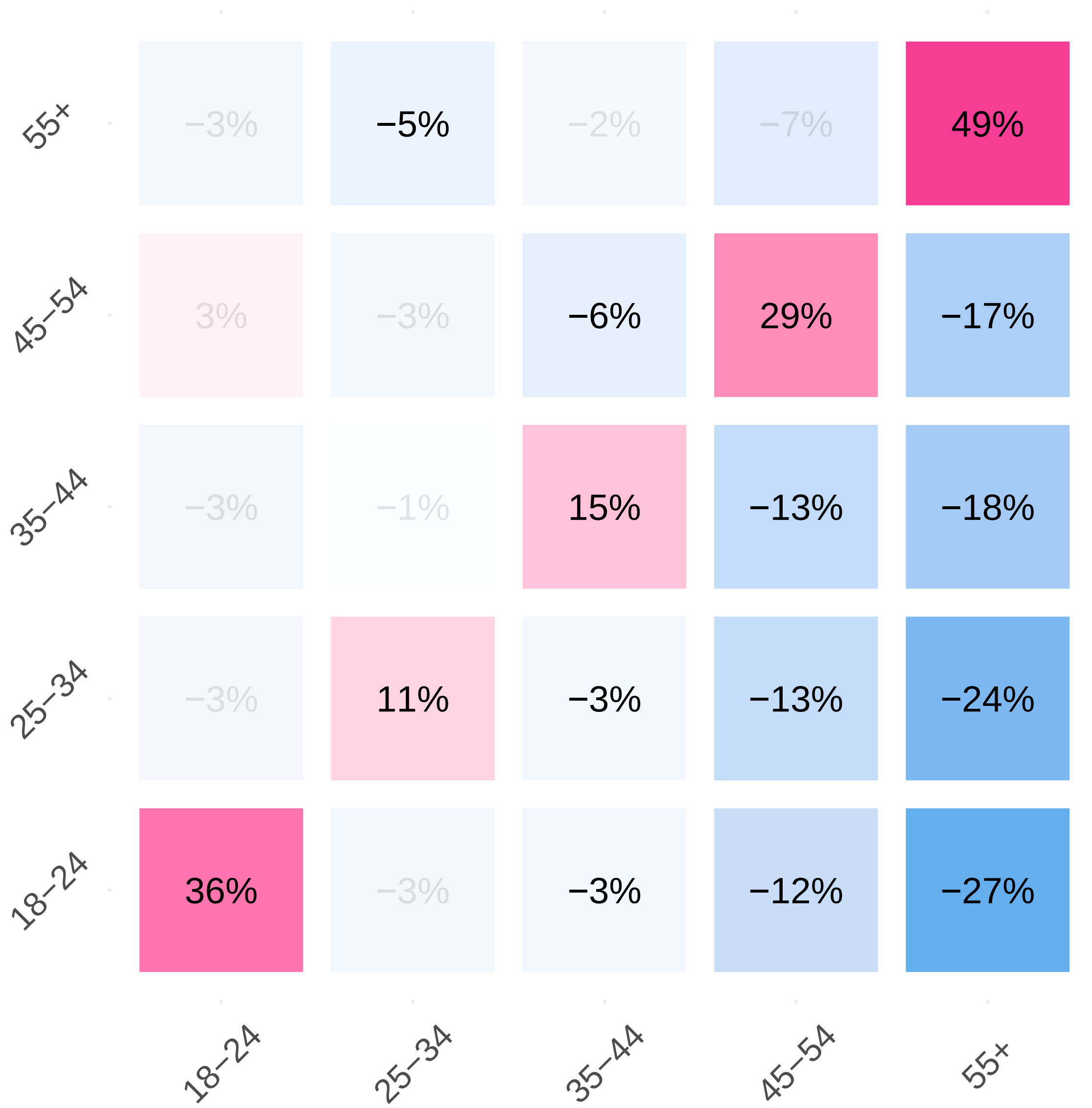


Figure 10.3: Social mixing matrix for MPX NYC participants by age

11 Outbreak control

11.1 Spatial efficiency

Figure 11.1 shows which community districts would have to be immunized to immunize 33% of the population (Group A), and 66% of the population (Group A + Group B) under both the contact-neutralizing and movement-neutralizing intervention strategies.

Under the contact-neutralizing strategy, it would take five community districts to immunize 33% of the population. These are MN04, BK01, MN10, BK03, and QN01. To reach 66% of participants with the intervention, it would be necessary to immunize 13 community districts. The additional community districts would be MN12, BK04, MN02, BK08, MN07, MN09, BK02, and MN05.

Under the movement-neutralizing strategy, it would take six community districts to reach 33% of the population: MN04, BK01, MN05, BK04, MN02, and BK03. It would take 16 community districts to reach 66% of the population. The additional community districts would be: MN64, MN10, QN01, BK08, MN09, QN05, BK09, MN11, BK14, and BK07.

11.2 Spatial network impact

Figure 11.2 shows that nearly 100% of individuals are included in the largest connected component before any intervention. For the movement-neutralizing strategy, the proportion of the population included in the largest connected component declines more sharply than for the contact-neutralizing strategy as more community districts are immunized. In the contact-neutralizing strategy, the overall proportion of unvaccinated individuals declines more sharply than in the movement-neutralizing strategy. There is not a noticeable difference between the two strategies for the first 10 community districts to be immunized - differences between the two strategies start emerging once 10 community districts are immunized.

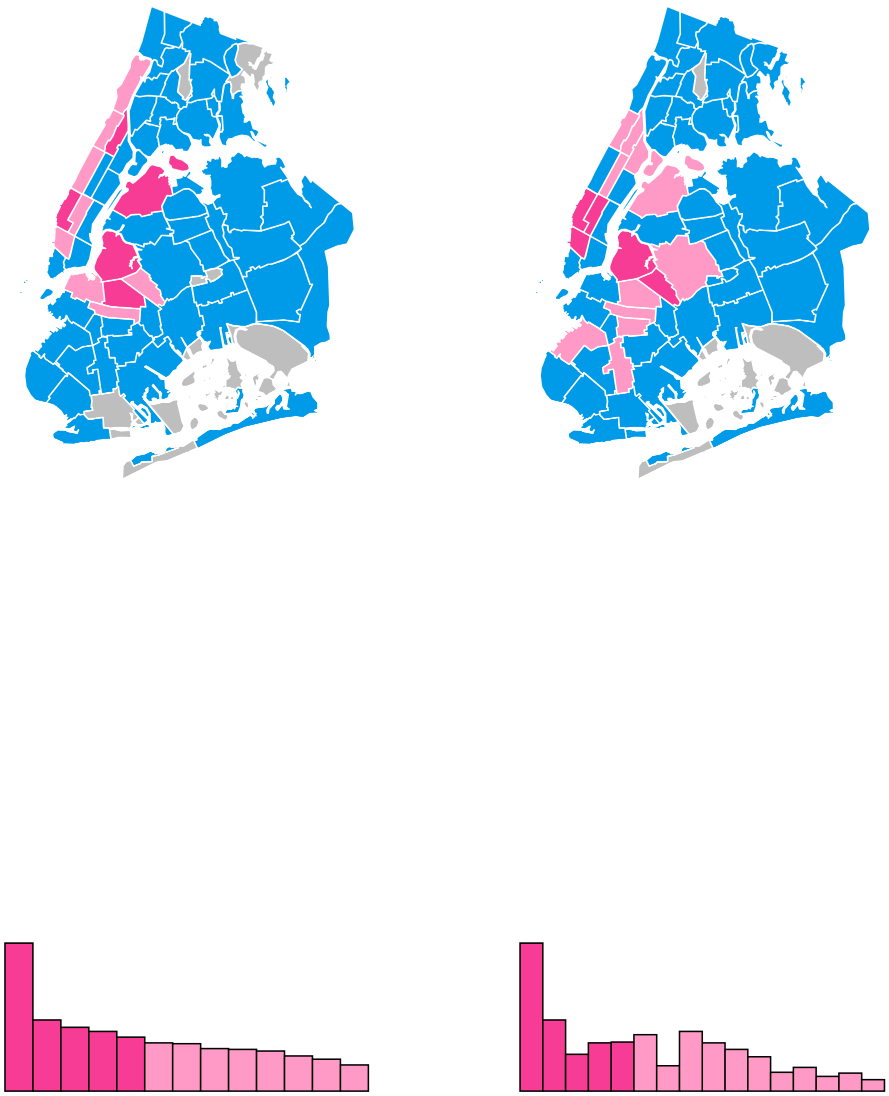


Figure 11.1: High-coverage community districts: contact vs movement-neutralizing strategy

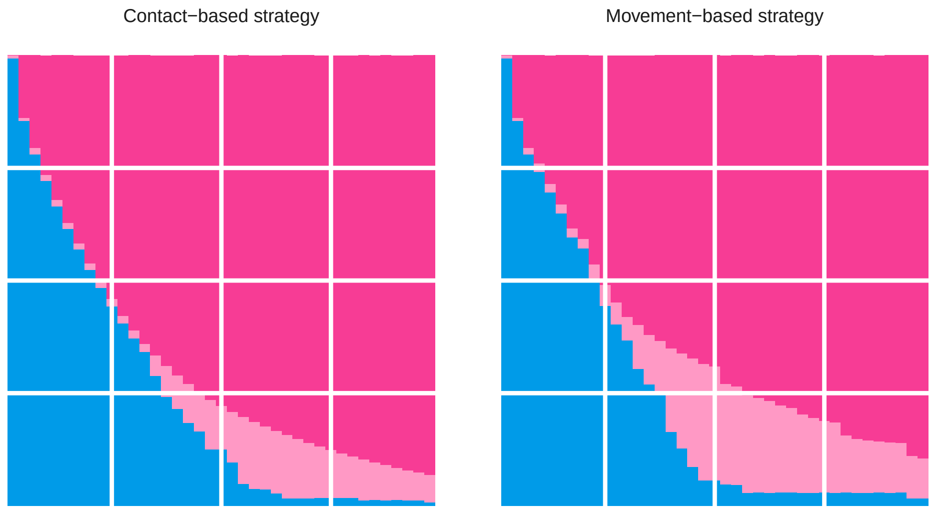
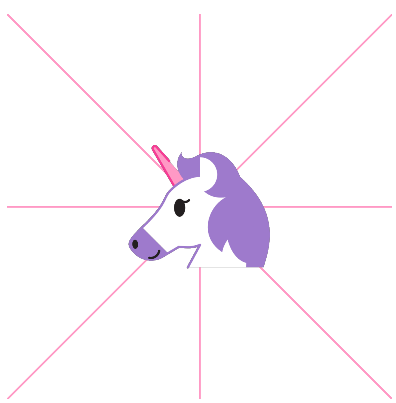


Figure 11.2: Network impact



Part IV

Conclusion

12 Discussion

13 Recommendations

Part V

Endnotes

14 References

Anna Grace Lee. n.d. "Will the Specter of Monkeypox Affect Pride Weekend in New York?" *New York Times*. https://www.nytimes.com/2022/06/25/style/monkeypox-nyc-pride.html.

Aronow, Peter M, and Cyrus Samii. 2017. "Estimating Average Causal Effects Under General Interference, with Application to a Social Network Experiment."

Blackburn, Dawn, Nicole M. Roth, Jeremy A. W. Gold, Leah Zilversmit Pao, Evelyn Olansky, Elizabeth A. Torrone, R. Paul McClung, et al. 2022. "Epidemiologic and Clinical Features of Mpox in Transgender and Gender-Diverse Adults - United States, May-November 2022." *MMWR. Morbidity and mortality weekly report* 71 (5152): 1605–9. https://doi.org/10.15585/mmwr.mm715152a1.

FCC. 2022. "Federal Communications Commission API." https://www.fcc.gov/reports-research/developers/api-terms-service.

"For Monkeypox Patients, Excruciating Symptoms and a Struggle for Care." 2022. *New York Times*, July. https://www.nytimes.com/2022/07/18/nyregion/new-york-monkeypox-vaccine.html.

Hernán, Miguel, and James M. Robins. 2018. *Causal inference*. Chapman & Hall/CRC monographs on statistics & applied probability. Chapman & Hall/CRC.

———. 2021. *Causal inference*. Chapman & Hall/CRC monographs on statistics & applied probability. Boca Raton: Chapman & Hall/CRC.

Krellenstein, James, Joseph Osmunson, and Keletso Makofane. 2023. "To Fight Monkeypox, Remember the Lessons of h.i.v." *New York Times*, March. https://www.nytimes.com/2022/05/29/opinion/monkeypox-covid-and-hiv.html.

Kyaw, Nang Thu Thu, Naama Kipperman, Karen A. Alroy, Jennifer Baumgartner, Addie Crawley, Eric Peterson, Amara Ross, et al. 2022. "Notes from the Field: Clinical and Epidemiologic Characteristics of Mpox Cases from the Initial Phase of the Outbreak - New York City, May 19-July 15, 2022." *MMWR. Morbidity and mortality weekly report* 71 (5152): 1631–33. https://doi.org/10.15585/mmwr.mm715152a3.

Makofane, Keletso. 2021. "Networked Resources and Health in South Africa: Towards a Materialist Social Network Epidemiology." PhD thesis, Cambridge, Mass. https://nrs.harvard.edu/URN-3:HUL.INSTREPOS:37370168.

Minhaj, Faisal S, Yasmin P Ogale, Florence Whitehill, Jordan Schultz, Mary Foote, Whitni Davidson, Christine M Hughes, et al. 2022. "Monkeypox Outbreak—nine States, May 2022." *Morbidity and Mortality Weekly Report* 71 (23): 764.

Neo4j. 2024. "Neo4j Documentation." https://neo4j.com/docs/?utm_source=Google&utm_medium=PaidSearch&utm_campaign=Evergreen&utm_content=AMS-Search-SEMCE-

DSA-None-SEM-SEM-NonABM&utm_term=&utm_adgroup=DSA&gad_source=1&gbraid=0AAAAADk9OYpn-veLtg3MSa3HV-mXEgnRC&gclid=Cj0KCQjwlvW2BhDyARIsADnIe-KKo4LZmo02jxyexPVweelrw_q-AwBtFSFPqVlfJ5YjiNyd1w7txfUaAk62EALw_wcB.

Robins, James. 1986. "A New Approach to Causal Inference in Mortality Studies with a Sustained Exposure Period—application to Control of the Healthy Worker Survivor Effect." *Mathematical Modelling* 7 (9-12): 13931512.

Roth, Grant, Jennifer Barnes-Balenciaga, Joseph Osmundson, Martez DR Smith, Nguyen K Tran, Nicholas Diamond, and Keletso Makofane. 2023. "Global North Learning from Global South: A Community-Led Response to Mpox in New York City." *PLOS Global Public Health* 3 (6): e0002042.

Tchetgen Tchetgen, Eric J, Isabel R Fulcher, and Ilya Shpitser. 2020. "Auto-g-Computation of Causal Effects on a Network." *Journal of the American Statistical Association*, 112.

US Census. 2022. "Census Tracts." https://www2.census.gov/geo/pdfs/education/CensusTracts.pdf.

Vercel. 2024. "NextJS Documentation." https://nextjs.org/docs.

15 About us

RESPND-MI is a collective of activists and researchers which formed to respond to the 2022 outbreak of MPOX among sexual and gender minority individuals in New York City.

[Lising of names and photos]

A Survey instrument

A.1 MPX NYC Person-Place Network Mapper

We designed, tested and implemented a survey tool we call the MPX NYC Person-Place Network Mapper. We named it this because the data collected through the instrument can be naturally organized into a bipartite network with nodes representing people and places, and edges representing relations between people and places. Organizing the data as a network allows the interpretation of network parameters such as node degree or connectedness as descriptors of the system connecting people to each other through the places they traverse within the city.

We designed the tool to solve practical and ethical issues related to the collection of sensitive spatial data in the context of an annonymous survey. Practically, we were interested in collecting data on locations participants visited and possibly had group sex in, but understood that we would not collect useful data if we collected markers like address or sip code using usual survey methods. In addition, we had hypothesized that a sizable proportion of sexual contact in a group happens in private settings whose addresses and exact locations participants were likely to forget.

Ethically, we were aimed to collect a data set that would limit the risk of identifying survey participants through the locations and other data disclosed through the survey. Collecting spatial coordinates would increase the risk of accidentally identifying people by, for instance, capturing the exact coordinate of a college dormitory as a participant's residence, or capturing the exact coordinates of a federal government office as a venue for sexual contact in a group setting. In both these examples, it becomes easier to narrow down the identity of participants using the location and the particiapnts' characteristics as recorded in survey data.

A.1.1 MPX NYC Person-Place Network Map

Though the investigator team doubted that participants would remember addresses or zip codes of places they visited, we thought that they might remember landmarks, cross-streets, neighborhood names, subway stations names, and other markers New Yorkers use to navigate the city so we used the Google maps interface and API.

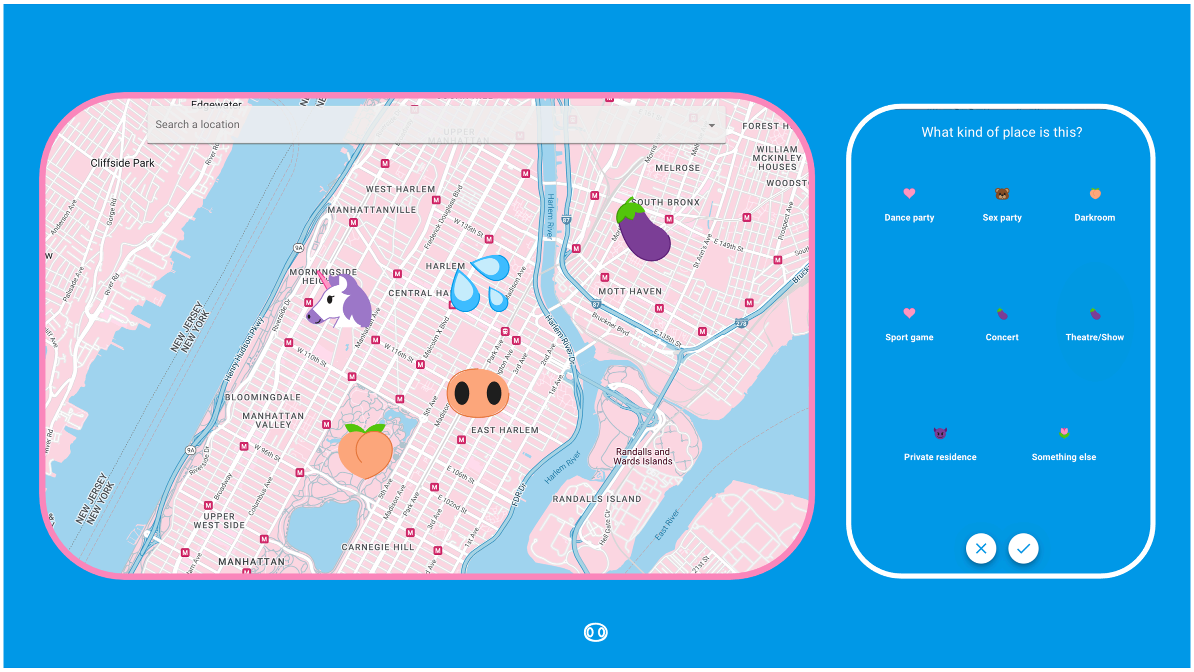


Figure A.1: Map input question format developed for MPX NYC

The MPX NYC Person-Place Network Map question displays a map with detailed geographic information including place names, roads, and neighborhood names. It includes a search bar as well as other navigation tools usually found on a google maps display (see Figure A.1).

To collect anonymous information on locations, first the participant is prompted to identify places of a certain type by tapping on the map. For instance, the first Network Map question asks participants to identify their primary residence. Once they select one, they are able to move onto the next question.

The Network Map also allows researchers to elicit a collection of locations. For instance, we ask participants to identify all the places they have had physical or sexual contact in a group setting. Each time the participant taps the map, a marker appears where they tapped, and a pop-up window asks questions about the most recently identified location. In our study, participants were asked if they had sexual contact at the location and to select the type of location it is from a closed list of options.

A.1.2 Coarsening to census tract

To ensure participant anonymity, the survey tool coarsens spatial data to the census tract level before sending them to the study database. Census tracts are non-overlapping spatial units that partition the United States into areas ranging from 2500 to 8000 in population size

(US Census 2022). This is done through The Federal Communications Commission API (FCC 2022) which receives longitude-latitude coordinates from the survey instrument, and returns census block identifiers, which are transformed to census tract identifiers by truncating the last X digits. [CITATION] It is only after spatial data are transformed that they are transmitted to the study database (backend).

A.1.3 Development

The front end of the MPX NYC Person-Place Network Mapper was developed as a single-page web application using the React/NextJS framework (Vercel 2024) with visual design based on the MPX NYC style guide. The back end was a custom-built neo4j graph database (Neo4j 2024). Along with the Network Map question format, the instrument allows the use of standard survey question formats (multiple choice, single choice, text field etc.).

A.1.4 Testing

The MPC NYC Person-Place Network Mapper underwent two rounds of user testing. About 10 people participated each time, returning feedback via an informal feedback questionnaire and via email.

B Community engagement

B.1 RESPND-MI Community forum

Starting in May 2022, we prepared a study about sexual activity and mpox symptoms among those most vulnerable to mpox. To promote consultation and collective decision making for the study, we organised an online community forum (CF). In the first CF, attendees expressed the need for a space to continually coordinate response efforts across organisations, and to address gaps not immediately met by local and federal public health agencies

CFs thus became weekly calls where participants received epidemiologic updates, discussed problems, and provided input on study design and marketing decisions. From June to September 2022, over 80 activists, community leaders, public health professionals participated in CFs. We produced policy briefs, op-eds, and educational materials; and conducted media outreach. The RST website became a clearing house for community-generated resources, as well as community-generated information on vaccination, testing, and treatment services.

Educational materials and policy recommendations from the CF were adopted by city, state, and federal PHAs. Reporters cited the RST in over 90 press clips. Survey data were used to inform NYC vaccination outreach services. (Roth et al. 2023).

The Community Forum shaped study plans, including the recruitment campaign, the design of a novel spatial data collection tool, and the development of our study questionnaire. Sustained engagement led to several concrete improvements in research approach. For example, a criticism of the initial communication approach was that it centered cisgender men to the exclusion of transgender women and transgender men. This was consistent with most communication about the social epidemiology of MPOX at the time, and most scientific reporting.

But given that many public heatlh data systems use measures for gender that are unable to identify transgender individuals, limiting attention to cisgender men would have limited our ability to understand the true extent of the mpox outbreak. We assumed that transgender people and cisgender people are part of the same social and sexual networks which would likely be affected by mpox. We invited the participant who raised that criticism to join the study team. Over time, we attempted to continually improve gender related language and iconography in the project through discussion among investigators.

B.2 MPX NYC marketing and communication

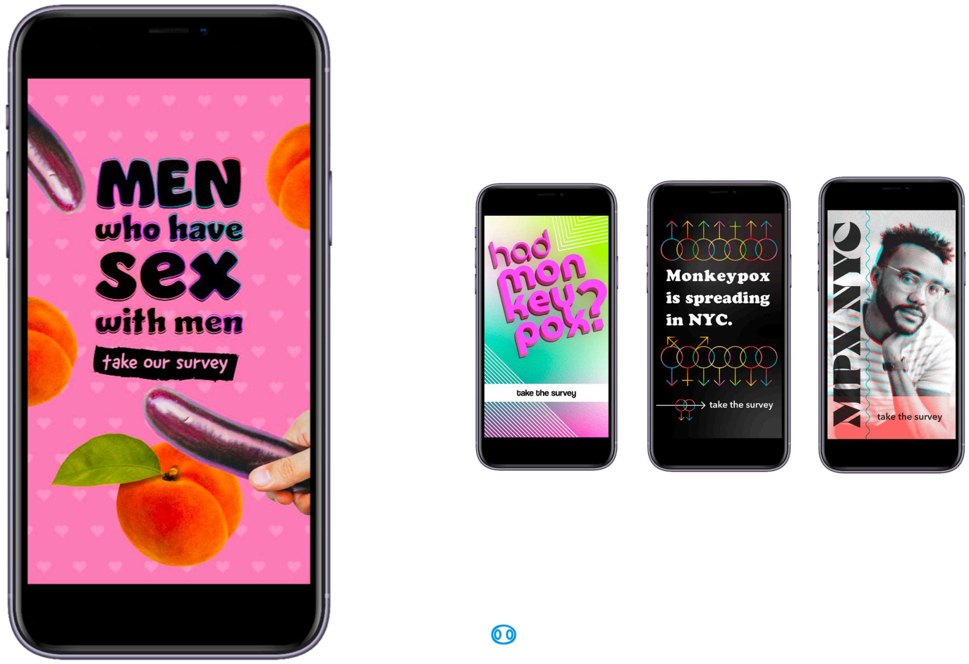


Figure B.1: Options for visual identity of MPX NYC"

To maximize the speed of recruitment, we invested in a bilingual, culturally competent communication campaign in collaboration with a marketing agency. The brief was to create a name and marketing concept that convey scientific rigor, build trust with the community, and stand out in a crowded online marketplace (see **?@fig-requirements**). As a starting point, the RESPND-MI team suggested a campaign that playfully uses emojis that have sexual connotations in queer and trans communities, and that integrates LGBT social media influencers as a communication channel.

Two campaign names and four marketing concepts (see Figure B.1) were developed by the marketing agency and presented to a meeting of the RESPND-MI Community Forum. The final concept and name were decided by consensus during that meeting.

Marketing materials were developed based on the concept. Materials included graphics, videos by queer influencers, and a style guide (including colors, icons, and fonts) for all public-facing materials of the MPX NYC study. We organized these materials into a "social media toolkit" and shared them through the study website to enable individuals and organizations to easily share them through their own social media accounts.

Participants were recruited via paid and donated advertising placed through Grindr, Twitter, Instagram, and outdoor electronic billboards.

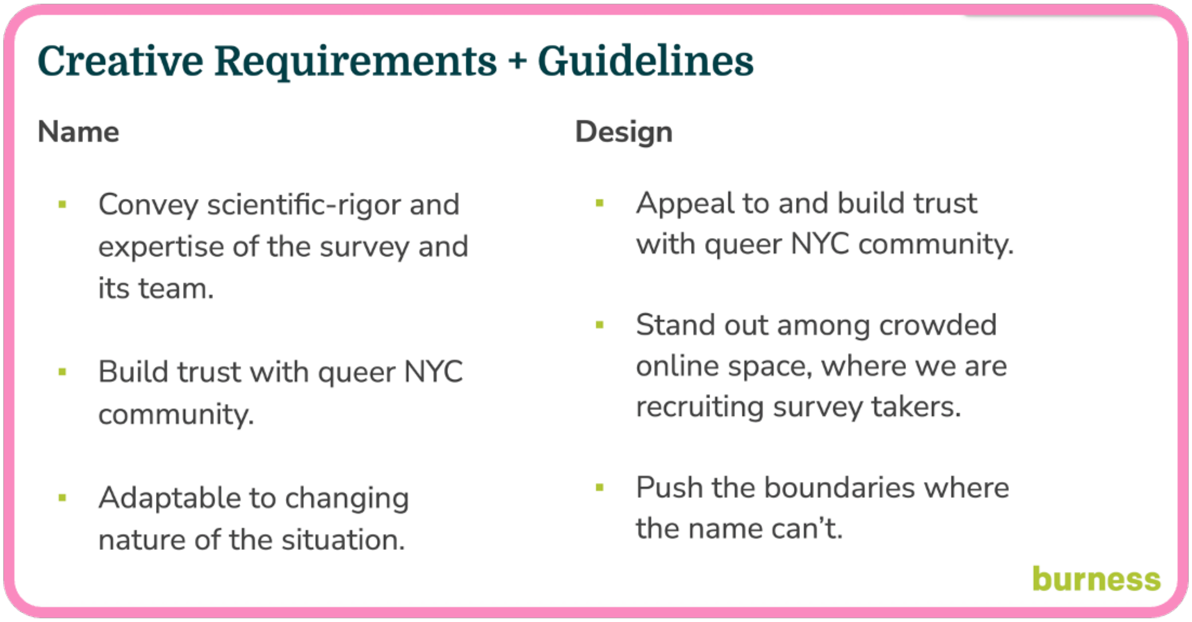


Figure B.2: Requirements for MPX NYC Study

C SSNAC I: Social & spatial

To motivate the Social and Spatial Network Analysis with Causal interpretation (SSNAC) framework, we consider a study investigating the risk of acquiring a cold among individuals (pictured as dots in physical space in each frame of Figure C.1) in a setting where there is one known infected person. The simplest approach would be to assign average risk of acquiring a cold to each individuals. The problem with this strategy, though, would be that not all individuals have the same risk once we know who the infected person is. Those who are more likely to interact closely with them have higher risk, but not all people are equally likely to interact.

A social network analysis would gain a more granular understanding of risk by assigning people who are connected to other people a higher risk of acquiring the cold than those who have no relationships. In addition, different connected components of the network could be assigned differnet levels of risk.

An ecologic approach would be to divide up the area into households and assign a different level of risk based on the household (third panel from left). If we knew who has a cold, we could assign people in the same household higher risk for acquiring the cold than others. This anlysis, however, would ignore the fact that households are not equally connected.

A spatial analysis would enhance this analysis by enriching household data with a spatial weights matrix indicating which pairs of households are neighrbors (second panel from right). Then the risk of aquiring a cold in a given household would be a function of the risk of aquiring a cold among individuals in neighboring households.

An SSNAC analysis (rightmost panel) would take advantage of both spatial and social network data, assigning risk so that the risk of acquiring the cold is a function of the household the individual belongs to, the neighboring household, and the social network connections.

C.1 SSNAC components

The SSNAC framework is an approach to collecting, analyzing and interpreting data which explicitly accounts for relationships among people, the places they physical places they interact in or co-inhabit.

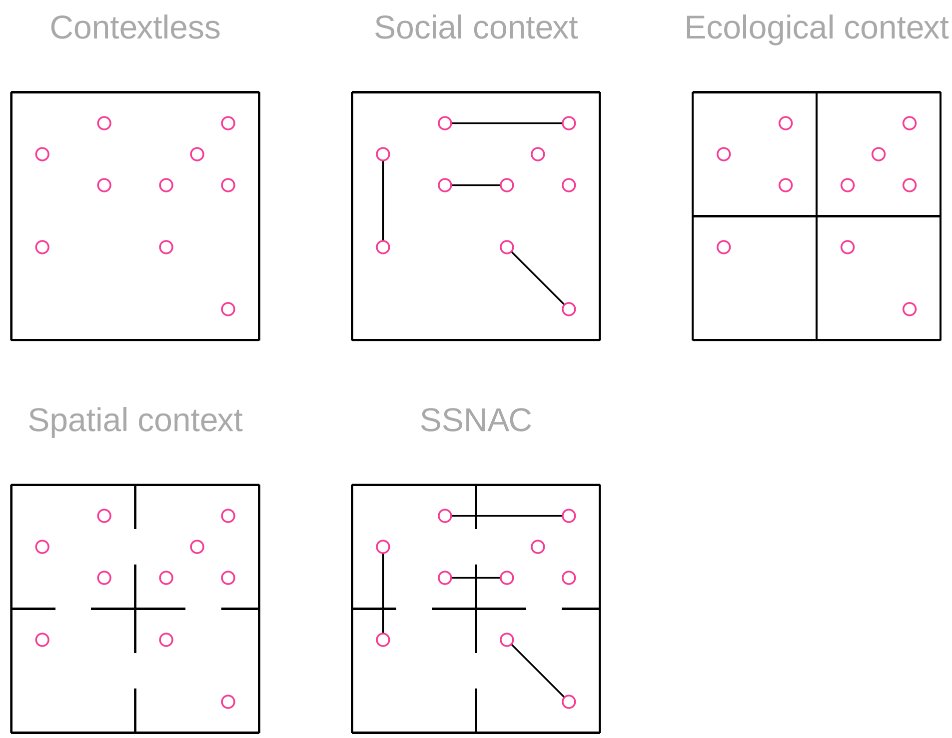


Figure C.1: Approaches to incorporating context in studies of individual health

It is applicable in two situations: when investigators want to describe the patterns in the relations among people, places, and across people and places; or when investigators are interested in inferring causality in the context of the system formed by people and places.

For the descriptive interest, it is necessary to have data on individuals, the physical spatial units they co-inhabit, relationships among individuals (network data), as well as the relationships among spatial units (spatial weights matrix).

For the causal interest, it is necessary for the investigator should have the data mentioned above, as well as a theory or set of theories about the causal process under investigation. These theories must explicitly state:

- the meaning of nodes (which entities they represent)
- the meaning of ties (how they relate the entities they link)
- an understanding of which ties are relevant to the causal question under investigation

C.2 The data are a network

The central data structure in a SSNAC is the person-place graph. Figure C.2 depicts the rightmost frame of Figure C.1 by showing spatial units as nodes in a network, connecting adjacent spatial units with a network edge, and connecting individuals with the households they belong to. To show the flexibility of this data object, we added an additional type of relationship an individual can have with a household - that of an employee.

The majority of study data are stored as node or edge attributes. For instance, person nodes might have attributes like age and race whereas place nodes could have attributes like density or surface area. An edge linking a person to a place might have an attribute conveying how the person relates to the place. For instance in Figure C.2, some edges indicate residence, some indicate next

C.3 The network is a causal structure

The Person-place graph is a starting point for constructing a causal model, but it is not sufficient. Depending on the causal question, some edges in this graph are relevant and some are not. For example, if we were interested in a rumor network, we might decide to depict only people and not places in the causal graph. We might also choose to ignore the neighbor edges and focus only on the friend and employee ones, if it is the case that rumours are transmitted within households and across friendships but not to neighbors.

If we were interested in an infectious disease, and in the study setting, there is a stay-at-home order so that friends do not meet, but neighbors possibly meet, then we might keep the

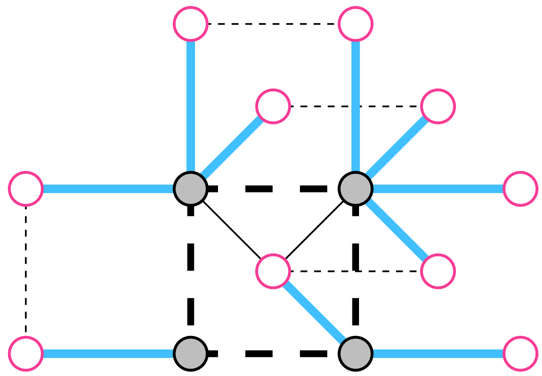


Figure C.2: SSNAC Person-place graph

neighbor ties and drop the friend and employee ties. In either case, it is necessary to make changes to the Person-place network through network operations like ignoring or adding ties, or even grouping nodes to form a different graph altogether.

We explore this in the Analysis appendix.

C.4 New causal structures open new causal enquiries

There are four types of causal queries that focus on node-level outcomes in the SNACC framework.

- Person-To-Place: Evaluating the causal impact of a person-level intervention on place-level outcomes for some defined set of people and some defined set of places. e.g. What is the impact of vaccinating college-aged adults in New York City on mpox incidence in Union Square (a favourite hangout spot for college students)
- Person-to-Person: Evaluating the impact of a person-level intervention on person-level outcomes for some defined sets of people. e.g. How does vaccinating all cisgender white men in New York City affect mpox incidence among Black and Latinx cisgender black men in New York City?

- Place-to-Place: Evaluating the impact of a place-level intervention on place-level outcomes for some defined sets of places. e.g How does vaccinating Manhattan affect mpox incidence in Brooklyn?
- Place-to-Person: Evaluating the impact of a place-level intervention on person-level outcomes for some defined set of people and some defined set of places. e.g. How does vaccinating Harlem affect mpox incidence among cisgender black men in New York City

In addition there is an endless potential for constructing causal queries that focus on characteristics that cannot be represented as node characteristics for any single node, but rather are characteristics that emerge as a result of the pattern of interactions among nodes. For examples, see the section on social causal inference.

D SSNAC II: Network analysis

In this appendix we describe the main data objects of the MPX NYC study - the Person-place graph, the Person graph and the Place graph. We define measures used to analyze these graphs.

D.1 Data

D.1.1 Person-place graph

The MPX NYC spatial participant data network is a network object that contains all collected data. In this network, nodes comprise of community districts and survey participants. Edges can only exist between a person and a communtiy district - they indicate that the person either lives in or has a contact place in that community district. In Figure D.1, participant nodes are blue and community district nodes are pink. Table D.1 (a) shows the node and Table D.1 (b) the edges of the network.

D.1.2 Person graph

The MPX NYC participant network has participants as nodes. A pair of nodes is connected by a weighted edge if each of the individuals represented by the nodes has at least one contact venue or residence in the same, shared community district. The weight of the edge is given by the count of community districts the pair of nodes have in common. By definition, an edge of weight 0 is equivalent to the absence of an edge. Table D.4 shows the nodes of the participant network shown in Table D.4 (a) and Table D.4 (b) show the edges.

D.1.3 Place graph

The MPX NYC community district network has community districts as nodes. A pair of community districts is connected by a weighted edge when at least one participant has a contact venue or residence in both community districts. The weight is the count of participants which are shared by the two community districts. Figure D.3 shows the community district network derived from the MPX NYC spatial participant data network shown in Figure D.1. Table D.7 (a) shows list the nodes of the former network, and Table D.7 (b) shows the edges.

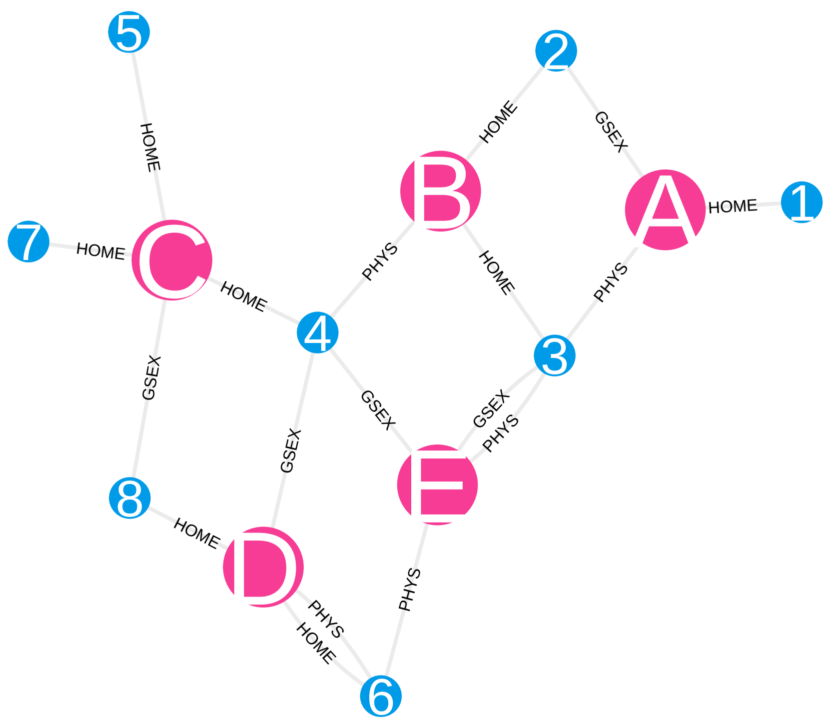


Figure D.1: Sample person-place network visualization

Table D.1: Sample person-place network data

(a) Nodes

Label	Age	Type
1	18-25	Person
2	18-25	Person
3	18-25	Person
4	18-25	Person
5	26-50	Person
6	26-50	Person
7	26-50	Person
8	26-50	Person
A	-	Community District
B	-	Community District
C	-	Community District
D	-	Community District
E	-	Community District

(a) Edges

From	To	Relation
1	A	HOME
2	A	GSEX
2	B	HOME
3	A	PHYS
3	B	HOME
3	E	GSEX
3	E	PHYS
4	B	PHYS
4	C	HOME
4	D	GSEX
4	E	GSEX
5	C	HOME
6	D	PHYS
6	D	HOME
6	E	PHYS
7	C	HOME
8	C	GSEX
8	D	HOME

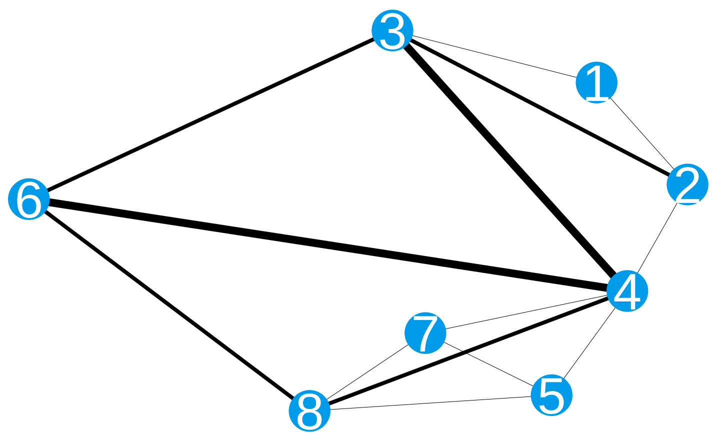


Figure D.2: Sample person-person network visualizatoin

Table D.4: Sample person-person network data

(a) Nodes

Name	Type
1	Person
2	Person
3	Person
4	Person
5	Person
6	Person
7	Person
8	Person

(a) Edges

From	To	Weight
1	2	1
1	3	1
2	3	2
2	4	1
3	4	3
3	6	2
4	5	1
4	6	3
4	7	1
4	8	2
5	7	1
5	8	1
6	8	2
7	8	1

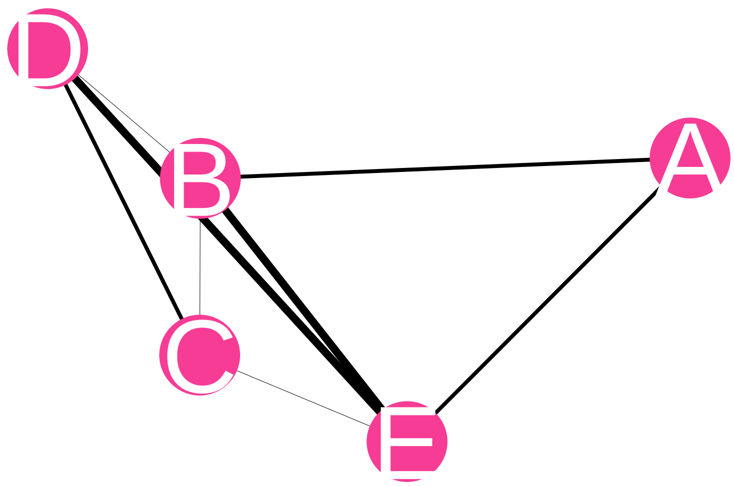


Figure D.3: Sample place network visualization

Table D.7: Sample place network data

(a) Nodes

Name	Type
A	Community District
B	Community District
C	Community District
D	Community District
E	Community District

(a) Edges

From	To	Weight
A	B	2
A	E	2
B	C	1
B	D	1
B	E	3
C	D	2
C	E	1
D	E	3

D.2 Measures

In this section, we describe network measures that are both useful statistics to interpret in describing MPX NYC data and useful intermediate steps in other measures we constructed.

We defined four measures related to node degree which offer a useful interpretation of MPX NYC data: participant social reach (R_i), participant spatial reach (S_i), community district social reach (r_j), and community district spatial reach (s_j). For each of these, we developed a measure for which we count only the connections to nodes which are included in some defined subset $A \subset N$ of nodes. We thus define conditional participant social reach ($R_i|A$), conditional participant spatial reach ($S_i|A$), conditional community district social reach ($r_j|A$), and conditional community district spatial reach ($s_j|A$).

We further defined two measures which summarize the pattern of connection across the people and places in the MPX NYC dataset. The first, the spatial mixing coefficient, can be used to analyze the pattern of connection across different demographic groups. The second, proportion in largest connected component, can be used to analyze the overall connectedness of the MPX NYC Person-Place graph and related graphs.

D.2.1 Notation

Define $\psi(i,j)$ as the adjacency function for the person-place graph. For person i and community j:

$$\psi(i,j) = \begin{cases} 1 & i \sim j \\ 0 & \text{otherwise} \end{cases}$$

Define $\psi_P(i,k)$ as the adjacency functon for the person graph. For persons i and k:

$$\psi_P(i,k) = \sum_{j \in N_C} \psi(i,j)\psi(k,j)$$

Define $\psi_C(j,l)$ as the adjacency functon for the place graph. For places j and l:

$$\psi_C(j,l) = \sum_{i \in N_P} \psi(i,j)\psi(i,l)$$

D.2.2 Node characteristics

D.2.2.1 Participant social reach

Participant social reach is defined as weighted node degree in the MPX NYC participant network. For participant $i \in N_p$

$$R_i = \sum_{j \in N_p} \psi_p(i,j)$$

$$R_i|A = \sum_{j \in N_p} \psi_p(i,j) I_{x \in A}(x = j)$$

where the indicator function $I_{x \in A}(x)$ equals 1 when $x \in A$ and 0 otherwise.

D.2.2.2 Participant spatial reach

Participant spatial reach is defined as the number of connections the focal participant has with community districts. It is defined as the weighted node degree in the MPX NYC spatial participant network. For $i \in N_p$

$$S_i = \sum_{j \in N_c} \psi(i,j)$$

$$S_i|A = \sum_{j \in N_c} \psi(i,j) I_{x \in A}(x = j)$$

D.2.2.2.1 Community district social reach

Community district social reach is defined as the number of connections the focal community district has with participants. It is defined as the weighted node degree in the MPX NYC spatial participant network. For $j \in N_c$:

$$r_j = \sum_{i \in N_p} \psi(i,j)$$

$$r_j|A = \sum_{i \in N_p} \psi(i,j) I_{x \in A}(x = i)$$

D.2.2.2.2 Community district spatial reach

Community district spatial reach is defined as the weighted node degree in the MPX NYC community district network. For $j \in N_c$

$$s_j = \sum_{i \in N_c} \psi_c(i, j)$$

$$s_j | A = \sum_{i \in N_c} \psi_c(i, j) I_{x \in A}(x = i)$$

D.2.3 Network characteristics

D.2.3.1 Movement matrix

Define $d^H(i)$ as the home district of person i. The movement coefficient $m(k, l)$ is defined as the count of connections by individuals i who live in home community district k with community districts l.

$$m(k, l) = \sum_{i \in \mathcal{N}_P} \sum_{j \in \mathcal{N}_C} \psi_C(d^H(i), j) I_{d^H(x)=k}(i) I_{x=l}(j)$$

The movement matrix is a square matrix whose ij^{th} entry is $m(i, j)$.

D.2.3.2 Spatial concentration chart

The spatial concentration proportion $c(k)$ quantifies the degree to which a community district is over-represented among destination community districts compared to home districts in the movement matrix. It is the difference of the margins of the movement matrix. For community district k

$$c(k) = \frac{\sum_l m(l, k) - m(k, l)}{\sum_{i,j} m(i, j)}$$

The spatial concentration chart plots teh spatial concentration proprotion for each community district.

D.2.3.3 Spatial mixing matrix

The MPX NYC participant network can also be used to analyze the pattern of connection across different demographic groups. We did this using the spatial mixing coefficient, which we motivate and define below.

Say there are three types of retired university professors in Boston - A) those who choose to live and socialize in neighborhoods with universities (where they can meet old colleagues and students); B) those who choose to live and socialize away from former colleagues and students; and C) those who choose neighborhoods without regard for university life. And say the proportion of people who are under 25 years of age in Boston is 20% and that the majority of this group are college students. People who are under 25 years of age would be over-represented in neighborhoods with college campuses and residences, and under-represented in those without.

To understand what kind of retired professor a particular person is, you could ask her of all the peopple she encountered in the past 12 months, what proportion were under 25. If she is of type C, we would expect her answer to be about 20%. If she avoids universities (i.e. is of type B), the proportion would be lower than 20% and if she seeks them out (i.e. type A), her answer would be greater than 20%.

In the motivating example, we were soliciting the retired professor's preference for the subgroup: under 25. We were then mentally comparing it with the prevalence of that subgroup (20%). The spatial mixing coefficient makes this comparison by dividing preference by prevalence and subtracting 1. It is zero when contact venues and residences for a given subgroup are distributed in space independently of another; positive if members of the former subgroup tend to share community districts with those of the latter; and negative if they tend to not share community districts.

The spatial mixing coefficient is related to notions in social network analysis and compartmental infectious disease modelling. The coefficient of a given sub-group for that same subgroup indicates homophily when it is positive, random mixing when it is zero, and heterophily when it is negative. It goes further by showing, in the case of heterophily, which subgroups members of the focal subgroup tend to be connected with. This is useful in early outbreak investigations as it allows the analyst to understand the potential path of a new pathogen across the population of interest. e.g. where there is strong homophily in the a network like the MPX NYC participant network, a new pathogen will tend to stay within the subgroup it first affected. If there is strong heterophily, on the other hand, the pathogen will tend to disseminate across groups favored by the first subgroup it affected.

We define preference of group A for group B ($\Phi(A, B)$) as the average proportion of alters from group B among egos from group A.

$$\Phi(A, B) = \frac{1}{|A|} \sum_{i \in A} \frac{R_i|B}{R_i}$$

To understand whether preference of a specific group by another group is higher or lower than we would expect if ego's chose alters randomly, we compare preference for B to the prevalence of group B:

$$\frac{|B|}{|N_p|}$$

The spatial mixing coefficient is the ratio of preference to prevalence minus one:

$$\phi(A,B) = \frac{|N_p|}{|B|}\Phi(A,B) - 1$$

D.2.3.4 Proportion in largest connected component

In a given network, a connected component is a subset of nodes such that each node in the subset is reachable from any other node in the subset, and such that there are no nodes outside the subset which are reachable from a node inside it. By reachable, we mean possible to reach by tracing a path from one node to another over existing network ties.

In the MPX NYC participant network, it is possible to uniquely partition the node set N_p into connected components. This follows from the facts that connected components form non-overlapping subsets of N_p and that it is possible to write N_p as a union of subsets C_k which are each connected components.

$$N = \bigcup_{k=1}^{K} C_k$$

We define the proportion in the largest connected component as:

$$\theta_p = \frac{\max_k |C_k|}{|N_p|}$$

E SSNAC III: Causal interpretation

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RESPND-MI Collaboration

In this appendix we contextualize the main analysis in the MPX NYC project using a framework we name Social and Spatial Network Analysis with Causal interpretation (SNACC). The SSNAC framework can be used to guide the collection, management, analysis, and causal interpretation of network data. It offers guidance on conceptualizing and organizing spatial and social data as networks (Appendix C) and provides tools for descriptive and causal analysis (Appendix D).

In this section we elaborate the "causal interpretation" aspect of SSNAC. We define SSNAC as a set of straight-forward extensions of the currently dominant framework for conducting causal inference with observational data in epidemiology. We begin by giving an overview of the dominant approach, which we will call "classic causal inference". We extend this system to situations where data are characterized by interdependence of observational units rather than independence, naming this system "network causal inference". Finally, we accommodate social and spatial data simultaneously in the SSNAC framework. In each case, we describe the assumptions and implications of each system, illustrating with worked examples.

Classic causal inference, as articulated by Robins (1986), Hernán (2018), and others helps us to sharpen epidemiologic reasoning by providing a template for conducting thought experiments about population health. Typically in these thought experiments, the researcher reasons about the likely effect of some hypothetical individual-level intervention on some hypothetical outcome in some hypothetical population. Through transparent statistical assumptions, the quantities of interest in the thought experiment are calculated as a function of observed or observable data.

For instance, consider a hypothetical intervention where every person in the United States receives free, unlimited access to healthcare. Under classic causal inference, we can ask whether the prevalence of diabetes would change if we implemented the intervention. We can even ask about the extent of change and estimate that using data that have already been observed. The classic causal inference framework allows this by:

- making causal assumptions about the relationship between healthcare access and diabetes prevalence
- mathematically stating the hypothetical quantities we are interested in

- estimating those quantities using data we have or can collect about access to healthcare, diabetes prevalence, and any factor that determine both access and prevalence

A strong assumption made almost universally by practitioners of classic causal inference is that of independence between observational units or groups of observational units. In our opening example, this assumption holds that one person's access to healthcare does not affect any other person's diabetes status. It would imply that the diabetes status for a married man who does not know how to cook, for instance, would not at all be affected by improvements in health literacy experienced by his wife who cooks for the household. The assumption of independence runs counter to our intuitions about social life. Humans are networked - our lives are defined by our relationships with others.

The choice of adopting the independence assumption or not has profound implications for how we think about our inferential goals. In the frequentist tradition of statistics, the engine of uncertainty is repetition. We imagine that an experiment could be repeated over and over again and we summarize its behavior across those imagined repetitions. We do not often consider what exactly we mean by experiment and by repetition, though we make this distinction each time we decide how to analyse repeated-measures data.

An example to clarify: Consider a situation with two people, Jose and Thabo, who each toss the same coin once at time 1 and again at time 2. There are several ways of parsing this situation into "experiment" and "repetition". We could say that we are conducting:

- one experiment called "coin toss" and repeating it four times
- two experiments called "Thabo tosses a coin" and "Tumi tosses a coin" and repeating each experiment twice
- two experiments called "coin toss at time 1" and "coin toss at time 2" and repeating each experiment twice
- four experiments called "Thabo tosses a coin at time 1", "Thabo tosses a coin at time 2", "Tumi tosses a coin at time 1", and "Tumi tosses a coin at time 2 and not repeating any of them.

The choice between these interpretations turns on the experimenters prior beliefs about coin tossing: Does the likelihood of a head or tail change over time? Does it change based on the person tossing the coin? This kind of consideration is at the core statistical modelling, particularly when analyzing longitudinal data.

One consideration researchers do not often make is whether one person's coin toss is correlated with the other's. Consider the experiment "Thabo tosses a coin, then Tumi tosses a coin, then Thabo tossess a con, then Tumi tosses a coin". Whereas in the above listing, by virture of calling them separate experiments, we implicitly assumed that each coin toss was self-contained, in this latest iteration, we accommodate the possibility that the result of Thabo's coin toss affects the result of Tumi's.

We can quantitatively assess this possibility by repeating this sequence of coin tosses over and over, using that data to calculate a correlation coefficient between the outcomes of Thabo and Tumi's coin tosses. Without stating the experiment as this sequence, we have no framework with which to ask whether the Thabo and Tumi's coin tosses are related.

In epidemiologic practice, the modelling decision to categorize people or groups of people as independent of each other is so well-entrenched that it is often represented as a minimum condition of doing any valid causal quantitative research. Unlike coins, however, there is a mountain of evidence and experience showing that interdependence is the rule rather than the exception as far as people are concerned.

Our goal in this appendix and project is to offer alternative ideas and tools for causal inference that allow us to learn from interdependence rather than avoid it in pursuit of clear causal reasoning. We can have both clear reasoning, and an understanding enriched by the analysis of interdependence.

In crude terms, our approach involves moving from a framework where, as a rule, we consider the coin toss example one experiment, rather than break it up into four. i.e. Rather than define a scalar random variable to represent a measure made on n individuals, we will define a vector-valued random variable of length n and assume that some of the entries in that random variable are correlated as a result of being part of the same causal structure. But why would a researcher want to do that when asymptotic theory requires us to have as many repetitions as possible for valid statistical inference? How can we conduct valid inference and quantify uncertainty when we always have a sample size of 1?

Briefly, we can avail ourselves of asymptotic theory by making the assumption of no-spillover between observations, which is a weaker version of the independence between observational units assumption. No spillover means that one person's outcome can not directly affect another person's outcome. We further assume that the causal relationships between and within individuals are homogeneous in some way, allowing our estimates to benefit from having more observational units, even if they collectively comprise one realization of the statistical model. Under some reasonable assumptions, asymptotic theory can be used to improve estimates in a growing vector of inter-related observations, whereas it is usually applied to a growing collection of vectors of equal length. In the network causal inference and SSNAC section, we construct the boundedness restriction in the definitions of these models which help to limit how quickly the number of potential outcomes to estimate grows in relation to the number of observational units.

In the sections that follow, we describe classic causal inference, network causal inference, and the SSNAC framework, each time discussing the assumptions, causal model, causal estimands, and illustrating a causal identification strategy for the framework. Finally, we describe the MPX NYC outbreak response analysis presented in the main report using the SSNAC framework.

E.1 Classic causal inference

Example 1: Dress and Jobs

We are interested in settling an argument among friends about the relationship between dressing in formal wear and job offers. One friend thinks that dressing formally brings more job offers; everyone she knows who dresses formally has many more job offers than people who dress casually. The other friend thinks that what really brings job offers is currently having a job. In their reasoning, having a job causes people to dress formally and also causes people to receive job offers, but dressing formally does not in itself improve a person's likelihood of receiving job offers. We conduct a study to investigate.

We recruit three people, *Person 1*, *Person 2*, and *Person 3*. *Person 1* and *Person 2* are married to each other. *Person 1* and *Person 3* are newly acquainted colleagues. Say *Person 1* has job status L_1, *Person 2* has job status L_2 and *Person 3* has job status L_3. Each of these variables can only take on two values: "employed vs. unemployed". Similarly we define A_i as *Person i's* dress (formal vs. informal), and Y_i as their job offers (many vs. few). The job status measurement (L_i) precedes dress (A_i) in time, and dress precedes job offers (Y_i).

NPSEM-IE

Before we continue with the example, we give a brief overview of non-parametric structural equation models with independent errors (NPSEM-IE). This model underpins the causal inference practice of using counterfactual variables and using directed acyclic graphs (DAGs).

An NPSEM-IE is a system of equations which relates all the entries in a given vector-valued random variable with each other. It does so without specifying parametric functions for each equation in the system. An example of such a model is shown by Equation E.1. In this set of equations, values that are produced by equations higher up in the list are the inputs of equations lower down on the list. The functions f_* are deterministic, though their form is not necessarily known. The arguments U_* represent error terms - random variables which are independent of each other within and across observational units.

Whereas parametric models, for instance, a linear regression model with normally distributed errors, are characterized by their probability density functions, non-parametric models of this type are usually characterized by their conditional independencies. An example of a conditional independence statement would be:

for a realization $[X_1, X_2, X_3, X_4, X_5]$ *of some statistical model* F,

$$X_1 \perp X_3 | X_2, X_4$$

Non-parametric structural equation models are usually represented using the Directed Acyclic Graph (DAG), which show the relationships between the structural equations in the model. The DAG shown in Figure E.1, for instance, represents the model defined by Equation E.1. DAGs encode the conditional independencies that characterize non-parametric structural equation models. Using a DAG and a set of rules called d-separation (Hernán and Robins 2021), it is possible to read from the DAG the set of conditional independencies that imply all the conditional independencies that characterize the model.

Conditional indpendencies are indicated by missing edges in the DAG. If we made no assumptions at all about the causal process in Example 1 except the time-ordering of measurements, we could represent our data using Equation E.1. Since there are no missing edges between nodes in the Figure E.1, the structural equation model represents any possible joint distribution between the variables in the DAG - there are no conditional independencies.

$$\begin{aligned}
L_1 &= f_{L_1}(U_{L_1}) \\
L_2 &= f_{L_2}(U_{L_2}, L_1) \\
L_3 &= f_{L_3}(U_{L_3}, L_1, L_2) \\
A_1 &= f_{A_1}(U_{A_1}, L_1, L_2, L_3) \\
A_2 &= f_{A_2}(U_{A_2}, L_1, L_2, L_3, A_1) \\
A_3 &= f_{A_3}(U_{A_3}, L_1, L_2, L_3, A_1, A_2) \\
Y_1 &= f_{Y_1}(U_{Y_1}, L_1, L_2, L_3, A_1, A_2, A_3) \\
Y_2 &= f_{Y_2}(U_{Y_2}, L_1, L_2, L_3, A_1, A_2, A_3, Y_1) \\
Y_3 &= f_{Y_3}(U_{Y_3}, L_1, L_2, L_3, A_1, A_2, A_3, Y_1, Y_2)
\end{aligned} \tag{E.1}$$

But the most flexible model is not the most useful one. To fully characterize this system using observed data, we would need to estimate 9 functions using one observation of the 9-dimensional joint distribution that gave rise to the participants' observations. It would not help to recruit an additional person to the study because that would add three new equations to the model and we would have the problem of estimating 12 functions using one observation of a 12-dimensional joint distribution.

Epidemiologists take several approaches to to simplify this system of equations by making assumptions that leave us with fewer functions to estimate and more observations per function. They add restrictions to the model. In the next few sections, we will discuss the restrictions made under classic causal inference, network causal inference, and the SSNAC framework. In each case, simplifying the system of equations equates to deleting some of the arrows in the DAG shown in Figure E.1 (or equivalently, deleting some arguments from the functions in Equation E.1), or assuming that some of the functions in the system are identical to each other.

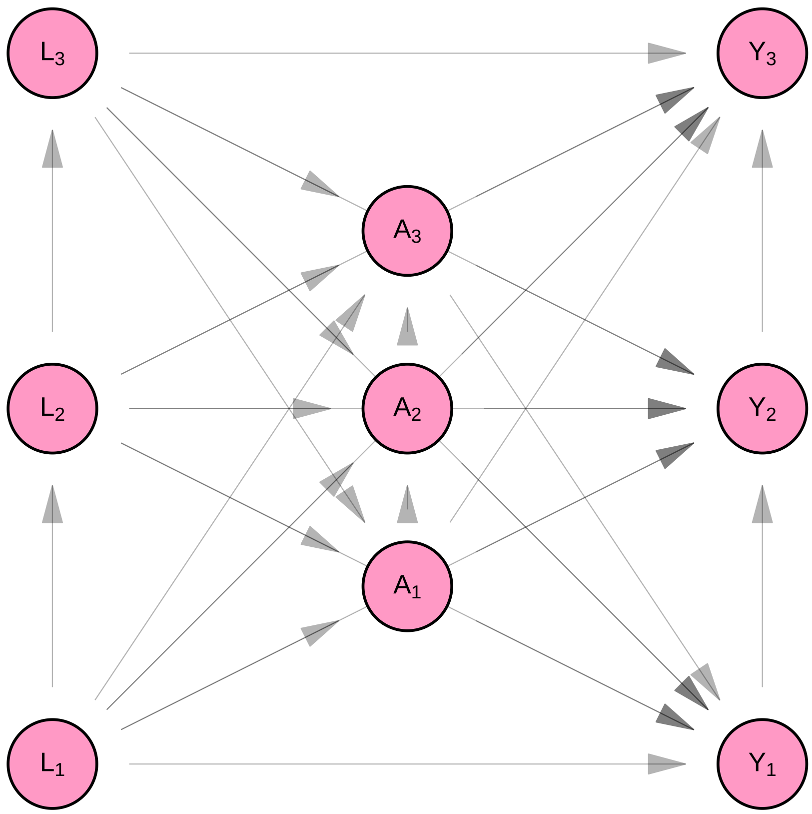

Figure E.1: Fully connected causal DAG

E.1.1 Assumptions

Under classic causal inference, we assume that observations are independent and identically distributed. We review these assumptions, naming them slightly differently for the purpose of comparing them with the assumptions made in subsequent sections. In each case, we divide assumptions up into *independence assumptions* (assumptions that delete arrows from the DAG) and *homogeneity assumptions* (assumptions that equate structural equations with each other).

E.1.1.1 Independence

E.1.1.1.1 No spillover

Under classic causal inference, we make the assumption of no spillover - that measurements taken across participants at the same time (e.g. L_1, L_2, and L_3) are independent of each other given all prior measures. i.e.

$$
\begin{gathered}
L_1 \perp L_2 \\
L_2 \perp L_3 \\
L_1 \perp L_3 \\[1em]
A_1 \perp A_2 | L_1, L_2, L_3 \\
A_2 \perp A_3 | L_1, L_2, L_3 \\
A_1 \perp A_3 | L_1, L_2, L_3 \\[1em]
Y_1 \perp Y_2 | L_1, L_2, L_3, A_1, A_2, A_3 \\
Y_2 \perp Y_3 | L_1, L_2, L_3, A_1, A_2, A_3 \\
Y_1 \perp Y_3 | L_1, L_2, L_3, A_1, A_2, A_3
\end{gathered}
$$

This is shown in Figure E.2, as the removal of arrows that connected the L_i's, A_i's and Y_i's. In Equation E.2, the L_i's are not functions of each other, the A_i's are not functions of each other, and the Y_i's are not functions of each other. This implies that given all the measurements taken prior to a set of contemporaneous measures, the contemporaneous measures are independent. This follows from the independence of the error terms U_{L_i}, the error terms U_{A_i}, and the terms U_{Y_i} across observations i.

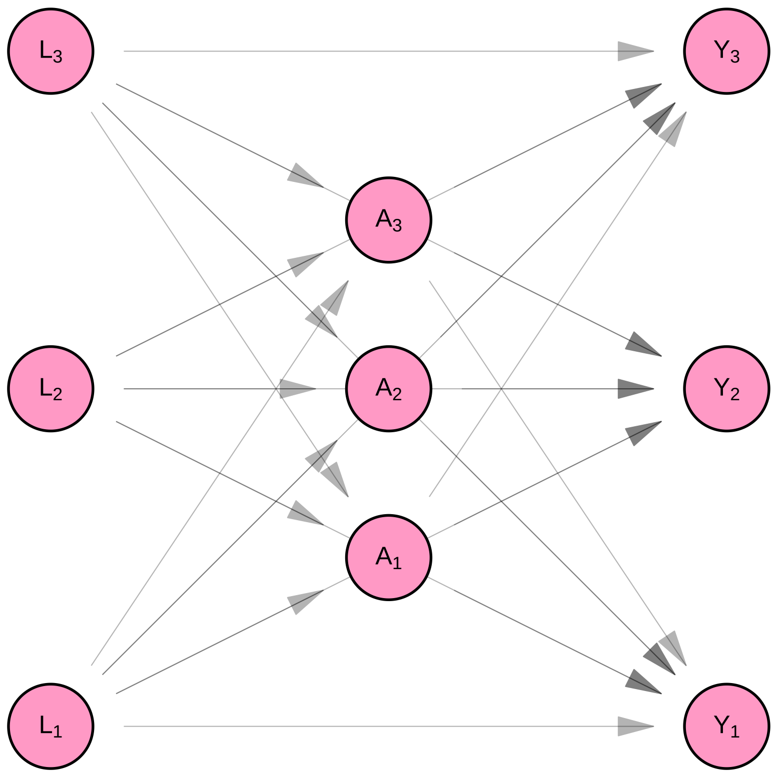

Figure E.2: Causal DAG with no spillover

$$
\begin{aligned}
L_1 &= f_{L_1}(U_{L_1}) \\
L_2 &= f_{L_2}(U_{L_2}) \\
L_3 &= f_{L_3}(U_{L_3}) \\
A_1 &= f_{A_1}(U_{A_1}, L_1, L_2, L_3) \\
A_2 &= f_{A_2}(U_{A_2}, L_1, L_2, L_3) \\
A_3 &= f_{A_3}(U_{A_3}, L_1, L_2, L_3) \\
Y_1 &= f_{Y_1}(U_{Y_1}, L_1, L_2, L_3, A_1, A_2, A_3) \\
Y_2 &= f_{Y_2}(U_{Y_2}, L_1, L_2, L_3, A_1, A_2, A_3) \\
Y_3 &= f_{Y_3}(U_{Y_3}, L_1, L_2, L_3, A_1, A_2, A_3)
\end{aligned} \tag{E.2}
$$

E.1.1.1.2 No interference

Classic causal inference makes the no interference assumption. We assume that measurements made across participants, but not at the same time are independent of each other. This is shown as the absence of arrows that connect L_i's to A_j's, L_i's to Y_j's and A_i's to Y_j for participants i, j with $i \neq j$. (See Figure E.3)

This allows us to simplify the structural equation model so that each participants measures are a function of only measures taken on them.

$$
\begin{aligned}
L_1 &= f_{L_1}(U_{L_1}) \\
L_2 &= f_{L_2}(U_{L_2}) \\
L_3 &= f_{L_3}(U_{L_3}) \\
A_1 &= f_{A_1}(U_{A_1}, L_1) \\
A_2 &= f_{A_2}(U_{A_2}, L_2) \\
A_3 &= f_{A_3}(U_{A_3}, L_3) \\
Y_1 &= f_{Y_1}(U_{Y_1}, L_1, A_1) \\
Y_2 &= f_{Y_2}(U_{Y_2}, L_2, A_2) \\
Y_3 &= f_{Y_3}(U_{Y_3}, L_3, A_3)
\end{aligned}
$$

E.1.1.2 Homogeneity

E.1.1.2.1 Identical distribution

Finally, we see above that the functions that generate observations L_i are similar in structure to each other, as are the the functions that generate A_i's and Y_is. We make the homogeneity assumption that the functions are identical to each other.

$$
\begin{aligned}
f_L &= f_{L_1} = f_{L_2} = f_{L_3} \\
f_A &= f_{A_1} = f_{A_2} = f_{A_3} \\
f_Y &= f_{Y_1} = f_{Y_2} = f_{Y_3}
\end{aligned}
$$

E.1.2 Model

E.1.2.1 Observed data

The homogeneity and indepdndence assumptions allow us to re-write the structural equation model dropping the person subscript from the equations. The model simplifies into one for three measures each taken on one person, rather than a model for 9 measures taken on 3 people. Since the measures on each person are identically distributed, this model can be used three times to separately generate the measures for *Person 1*, *Person 2*, and *Person 3*. By contrast, the original model produced measures for all three people at one go.

Figure E.3 and Equation E.3 show the resulting non-parametric structural equation model for our data, if we analyse it under classic causal inference. As a consequence of the assumptions stated in Section E.1.1, we can represent the model using three equations rather than 9. We retain the person subscript on the random variables in Figure E.3 and we keep all 9 nodes in the DAG on Equation E.3. This is to remind us that though our model is for one person, our purpose was always to infer something about the sample of three people.

$$
\begin{aligned}
L_i &= f_L(U_{L_i}) \\
A_i &= f_A(U_{A_i}, L_i) \\
Y_i &= f_Y(U_{Y_i}, L_i, A_i)
\end{aligned}
\tag{E.3}
$$

E.1.2.2 Counterfactual data

We define a counterfactual outcomes $Y_1^{(a_1)}$, $Y_2^{(a_2)}$, and $Y_3^{(a_3)}$, which represent the level of job opportunities, *Persons 1*, *2*, and *3*, would have if we set their dress to level a_1, a_2, and a_3, respectively. Then the model for counterfactual variables is given by Equation E.4.

It is represented using the Single World Intervention Graph (SWIG) pictured in Figure E.4. A SWIG represents a structural equation model where some of the equations are superseded by the value assigned by the experimenter. For so-called "treatment variables", the diagram separates the value of the variable as it would have been observed with no intervention (A_i) from

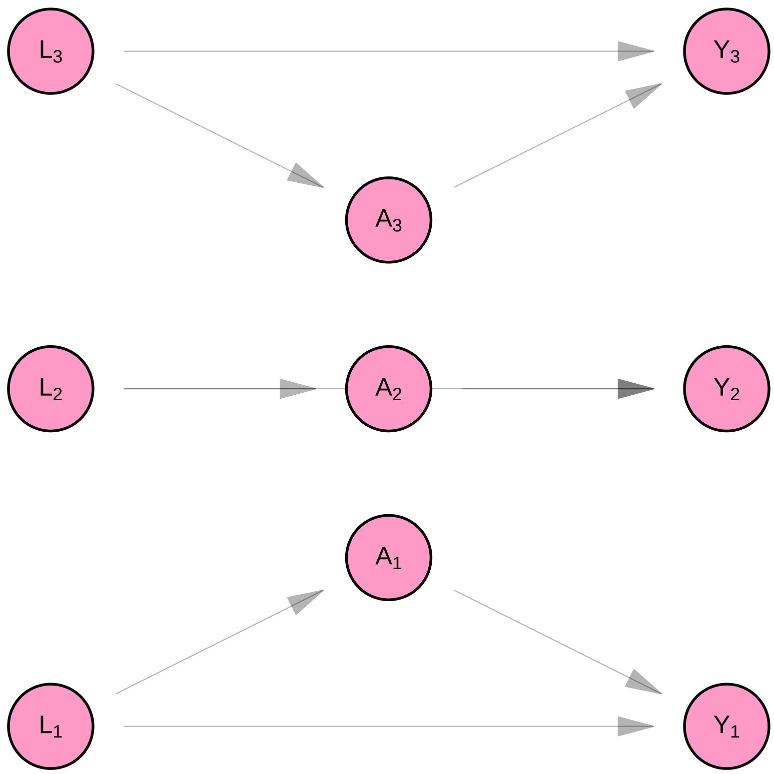

Figure E.3: Causal DAG with no interference or spillover the value that as it will be used in subsequent equations (a_i) as a result of the experimenter's intervention.

Note that because of the assumption of no interference and no spillover, each counterfactual outcome is a function only of the treatment value of the same individual.

$$\begin{aligned} L_i &= f_L(U_{L_i}) \\ A_i &= f_A(U_{A_i}, L_i) \\ Y_i^{(a_i)} &= f_Y(U_{Y_i}, L_i, a_i) \end{aligned} \tag{E.4}$$

E.1.3 Estimates

E.1.3.1 Estimands

We are interested in whether changing how formally someone dresses would change their level of job offers, on average, in our population. To determine the answer, we will compare the average counterfactual outcome of job opportunities for formal dress to the average counterfactual outcome of job opportunities for non-formal dress. i.e. We will make the following comparison.

$$E\left[\frac{1}{n}\sum_i Y_i^{(a=1)} - Y_i^{(a=0)}\right]$$

E.1.3.2 Identification

Once we have specified the hypothetical quantities from our thought experiment, we "identify" them, meaning write them in terms of the observed rather than counterfactual data. It is sufficient to identify the quantity:

$$E\left[\frac{1}{n}\sum_i E[Y_i^{(a)}|L_i]\right]$$

The inner expectation can be re-written as follows:

$$\begin{aligned} E[Y_i^{(a)}|L_i] &= E[Y_i^{(a)}|L_i, A_i = a] \\ &= E[f_Y(U_{Y_i}, l_i, a)|L_i, A_i = a] \\ &= E[Y_i|L_i, A_i = a,] \end{aligned}$$

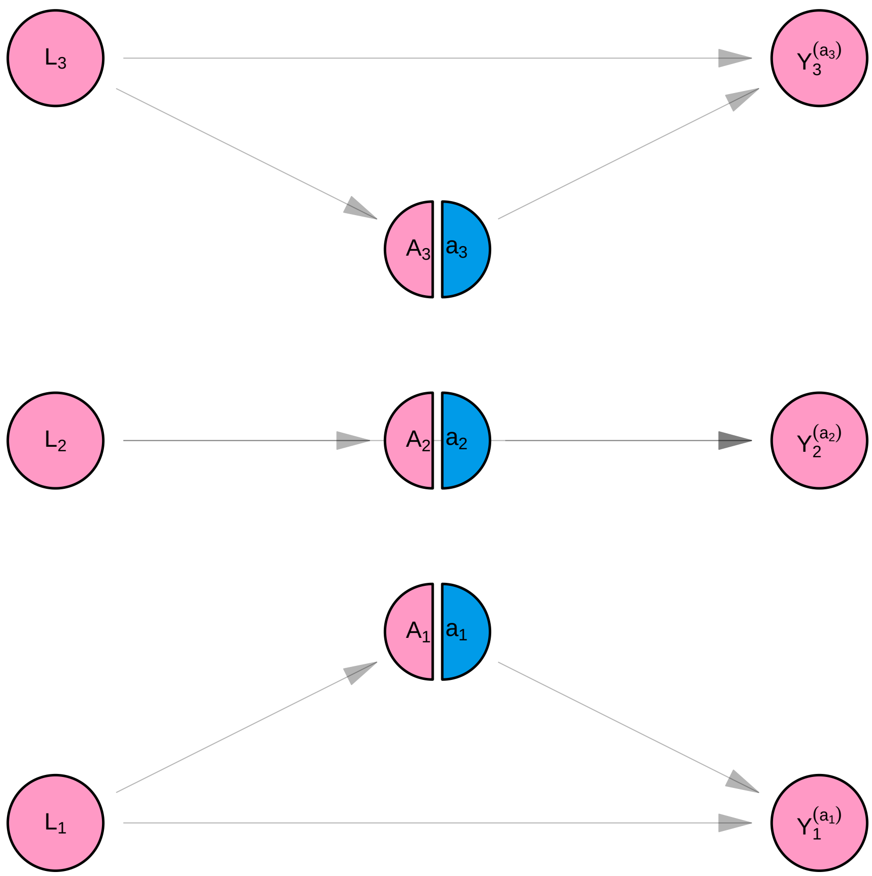

Figure E.4: SWIG with no interference or spillover

- In the first line, we use the fact that since it is constant, the value of dress set by the experimenter a is independent of the observed value of dress A_i for each person i.
- In the second line, we use the no spillover and no interference assumptions - person i's values (including counterfactual values) are independent of any measures not taken on person i.
- In the third line we substitute $Y_i^{(a)}$ for the structural equation that produces that variable, according to the model in Equation E.4.
- In the fourth line, we substitute the structural equation for the observed value Y_i according to the model in Equation E.3.

(Note that the function f_y in Equation E.4 is identical to the function f_y in Equation E.3. The error term for person i, U_{Y_i} has the same value in both the counterfactual and observational models, as does the value of l_i. As a result, $f_Y(U_{Y_i}, l_i, a)$ is equal to the observed value as well as the counterfactual value for participants i if $A_i = a$).

The functional we have to estimate from data is the following.

$$E[\frac{1}{n}\sum_i E[Y_i|L_i, A_i = a]]$$

We can evaluate the inner expectation (with respect to Y_i) by taking averages across sub groups in the study dataset. Then we have a function of random vector $\mathbf{L}$ to evaluate in the outer expectation.

$$\frac{\sum_i E[Y_i|L_i = l_i, A_i = a]I_{\{x=a\}}(a_i)}{\sum_i I_{\{x=a\}}(a_i)}$$

E.1.4 Discussion

E.1.4.1 Consistency

- Assumption that for a given individual observed and counterfactual distributions use the same structural equations and error term
- That means his counterfactual value under the treatment level he received is going to be equal to his observed value
- This is the assumption that allows us to substitute the observed value in the identification strategy above

E.1.4.2 Positivity

- Assumption that each combination of treatments and covariates has a non-zero chance of developing the outcome

E.1.4.3 Exchangeability

- We have not misspecified the DAG

Reflection questions

Using the structural equations:

1. Show that independence does not hold in the model defined by Equation E.1.
2. Show that independence holds in Equation E.3.
3. Show that the causal estimand in Section E.1 is equivalent to the usual definition of average causal effect (ACE)
4. Show that the causal estimate in Section E.1 is equivalent to the usual definition of the g-formula.
5. How is the SUTVA or consistency assumption encoded in this model?

E.2 Network causal inference

Example 2: Marriage, Dress, and Jobs

We mentioned that in our study sample that *Person 1* and *Person 2* are married to each other, and *Person 1* and *Person 3* are new colleagues. In the causal system connecting their job status, dress and job offers, an argument could be made that *Person 1*'s job status affects *Person 2*'s dress, since they pool their salaries as a married couple. It could also be argued that *Person 1*'s job status affects *Person 2*'s job opportunities if having a job brings the married couple into contact with other married couples who are connected to job markets. Finally, we could argue that *Person 1*'s dress affects *Person 2*'s job opportunities if some aspect of the reputation *Person 2* enjoys among potential colleagues derives from *Person 1*'s current job in that market. Symmetrically, *Person 2*'s current job could also potentially affect *Person 1's* job opportunities.

But what about the relationship between *Person 1* and *Person 3*? How does that shape the measurements of either person? We can imagine that the dress of a newly acquainted colleague would not affect the dress and job opportunities of her colleague. Whereas *Person*

1 and *Person 2* are causally related, *Person 1* and *Person 3* are not; their relationship is irrelevant to the causal system under investigation. *Person 2* and *Person 3* do not affect each other since they have no relationship at all.

Exposure mappings

In this section we give four definitions which are needed for the homogeneity assumptions of the Network Causal Inference model. Following Aronow and Samii (2017), we use the language of exposure mappings to express homogeneity assumptions of the structural equation model. Say we have a given measure $\mathbf{Y}$ whose entries are contemporaneous dichotomous measurements taken on n individuals. Say $\mathbf{Y}$ is part of an NPSEM which also includes dichotomous measures $\mathbf{X}_1$, $\mathbf{X}_2$, ..., $\mathbf{X}_k$ which were taken on all individuals prior to $\mathbf{Y}$.

Define $\mathcal{N}_i$ as the indices of all the observational units connected to unit i. Then the exposure mapping for Y_i as a function $g : \mathcal{M}_{n\times(k+1)} \to R$ such that:

$$Y_i = g(\mathbf{X}_1, \mathbf{X}_2, ..., \mathbf{X}_k, U_{Y_i})$$

E.2.0.1 Definition 1: Separability

We say an exposure mapping is separable if g can be written as follows:

$$Y_i = g(X_{1i}, X_{2i}, ..., X_{ki}, g_{1i}(\mathbf{X}_1, \mathcal{N}_i), g_{2i}(\mathbf{X}_2, \mathcal{N}_i), ..., g_{ki}(\mathbf{X}_k, \mathcal{N}_i), U_{Y_i})$$

where

$$g_{ki} : \mathbf{R}^n \times \Gamma \to \mathbf{R}$$

and Γ is the set of all alter sets $\mathcal{N}_i$.

$$\Gamma = \{\mathcal{N}_i\}_i$$

E.2.0.2 Definition 2: Equivalency

We say an exposure mapping g_{ti} is equivalent if for any individuals i, j with $i \neq j$:

$$g_{ti} = g_{tj} = g_t$$

In this case, we call g_t the network exposure mapping for $\mathbf{Y}$ with respect to $\mathbf{X}_t$ and we call $g_t(\mathbf{X}_{t\mathcal{N}_i})$ the network exposure value.

E.2.0.3 Definition 3: Symmetry

We say a network exposure mapping g_t is symmetric if for $\mathbf{X} \neq \mathbf{X}'$:

$$\mathbf{X}_{\mathcal{N}_i} = \mathbf{M}\mathbf{X}'_{\mathcal{N}_i} \implies g_t(\mathbf{X}, \mathcal{N}_i) = g_t(\mathbf{X}', \mathcal{N}_i)$$

For $|\mathcal{N}_i| \times |\mathcal{N}_i|$ permutation matrix $\mathbf{M}$ (square matrix whose row and column margins all equal 1).

i.e. the only entries in $\mathbf{X}$ which affect the value of $g_{ti}(\mathbf{X}, \mathcal{N}_i)$ are entries related to individuals $j \in \mathcal{N}_i$. The order of the entries does not matter.

E.2.0.4 Definition 4: Boundedness

We say that a network exposure mapping g_t is bounded if all possible network exposure values are contained by a finite set $\mathbf{D}$:

$$g_t(\mathbf{X}_t, \mathcal{N}_i) \in \mathbf{D}$$

Boundedness can obtain as a result of restrictions on the underliying network $\mathcal{N}_i$ or on the functional form of g_t. The former situation would arise if node degree is bounded. The latter can occur when we are modelling a network exposure which has a ceiling effect in its dose-response curve.

E.2.1 Assumptions

E.2.1.1 Independence assumptions

E.2.1.1.1 No spillover

Under network causal inference, we make the assumption of no spillover - that measurements taken across participants at the same time (e.g. L_1, L_2, and L_3) are independent of each other given all prior measures.

This is shown in Figure E.5, as the removal of arrows that connected the L_i's, A_i's and Y_i's in Figure E.1. In Equation E.5, the L_i's are not functions of each other, the A_i's are not functions of each other, and the Y_i's are not functions of each other. This implies that given all the measurements taken prior to a set of contemporaneous measures, the contemporaneous measures are independent.

(Note that it is not strictly necessary to make the no-spillover restriction to work with network data at all. Makofane (2021), following Tchetgen Tchetgen (2020), makes the assumption that contemporaneous measures arise from a conditional Markov random field. This complicates estimation in a way that is beyond the scope of this study. See **?@sec-ssnacx**.)

E.2.1.1.2 Network interference

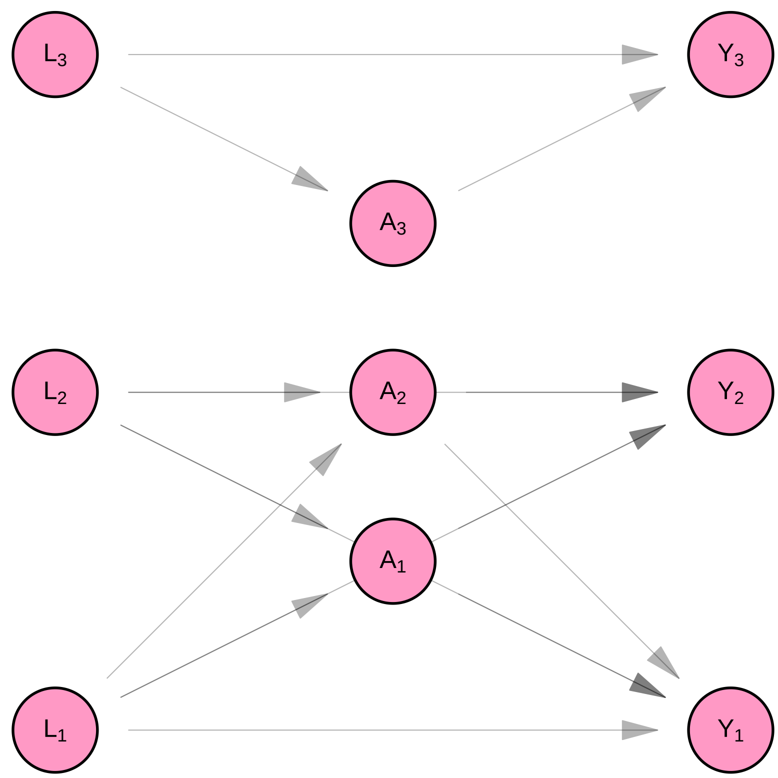

Figure E.5: Causal DAG with network interference

$$
\begin{aligned}
L_1 &= f_{L_1}(U_{L_1}) \\
L_2 &= f_{L_2}(U_{L_2}) \\
L_3 &= f_{L_3}(U_{L_3}) \\
A_1 &= f_{A_1}(U_{A_1}, L_1, L_2) \\
A_2 &= f_{A_2}(U_{A_2}, L_1, L_2) \\
A_3 &= f_{A_3}(U_{A_3}, L_3) \\
Y_1 &= f_{Y_1}(U_{Y_1}, L_1, L_2, A_1, A_2) \\
Y_2 &= f_{Y_2}(U_{Y_2}, L_1, L_2, A_1, A_2) \\
Y_3 &= f_{Y_3}(U_{Y_3}, L_3, A_3)
\end{aligned}
\tag{E.5}
$$

Under network causal inference, we make the assumption that interference respects some mapping between individuals - a mapping given by a network or set of networks. In Figure E.5, Based on their relationships with one another, *Person 1*'s measures are connected with *Person 2*'s, and neither are connected with P*erson 3*. See Aronow and Samii (2017).

E.2.1.2 Homogeneity assumptions

E.2.1.2.1 Identical distribution

As in the case of classical causal inference, we assume that the structural equations that produce each individual's observations are identical across individuals. i.e. we assume that for every individual i:

$$
\begin{aligned}
f_A &= f_{A_i} \\
f_L &= f_{L_i} \\
f_Y &= f_{Y_i}
\end{aligned}
$$

E.2.1.2.2 Homogenous interference

Separability

We assume that the exposure mapping for each variable in the structural model is separable. i.e. We define g_{A_i}, h_{Y_i}, and g_{Y_i} as network exposure mappings for A_i with respect to $\mathbf{L}$, for Y_i with respect to $\mathbf{A}$, and for Y_i with respect to $\mathbf{L}$. Seperability allows us to write the model as follows:

$$
\begin{aligned}
L_1 &= f_{L_1}(U_{L_1}) \\
L_2 &= f_{L_2}(U_{L_2}) \\
L_3 &= f_{L_3}(U_{L_3}) \\
A_1 &= f_{A_1}(U_{A_1}, L_1, g_{A_1}(\mathbf{L}_{\mathcal{N}_1})) \\
A_2 &= f_{A_2}(U_{A_2}, L_2, g_{A_2}(\mathbf{L}_{\mathcal{N}_2})) \\
A_3 &= f_{A_3}(U_{A_3}, L_3, g_{A_3}(\mathbf{L}_{\mathcal{N}_3})) \\
Y_1 &= f_{Y_1}(U_{Y_1}, L_1, A_1, g_{Y_1}(\mathbf{L}_{\mathcal{N}_1}), h_{Y_1}(\mathbf{A}_{\mathcal{N}_1})) \\
Y_2 &= f_{Y_2}(U_{Y_2}, L_2, A_2, g_{Y_2}(\mathbf{L}_{\mathcal{N}_2}), h_{Y_2}(\mathbf{A}_{\mathcal{N}_2})) \\
Y_3 &= f_{Y_3}(U_{Y_3}, L_3, A_3, g_{Y_2}(\mathbf{L}_{\mathcal{N}_3}), h_{Y_2}(\mathbf{A}_{\mathcal{N}_3}))
\end{aligned}
$$

Equivalency, symmetry, and boundedness

We assume that network exposure mappings g_A, h_y, and g_Y are equivalent, symmetric, and bounded. This

E.2.2 Model

E.2.2.1 Observed data

Figure E.6 and Equation E.6 show the resulting non-parametric structural equation model for our data, if we analyse it under network causal inference. Because of the assumption of identical distributions, we can represent the model using three equations rather than 9. We retain the person subscript on the random variables in Equation E.6 and we keep all 9 nodes in the DAG on Figure E.6 and add six additional nodes to represent network exposure values with respect to $\mathbf{A}$ and $\mathbf{L}$.

We draw the DAG like this to clarify that though our model is for one person, it represents a part of a system that relates all the people in the sample with each other. Our original purpose, after all, was to infer something about causality in the context of a given population.

We can re-write the structural equation model like this:

$$
\begin{aligned}
L_i &= f_L(U_{L_i}) \\
A_i &= f_A(U_{A_i}, L_i, g_A(\mathbf{L}_{N_i})) \\
Y_i &= f_Y(U_{Y_i}, L_i, g_Y(\mathbf{L}_{N_i}), A_i, h_Y(\mathbf{A}_{N_i}))
\end{aligned}
\tag{E.6}
$$

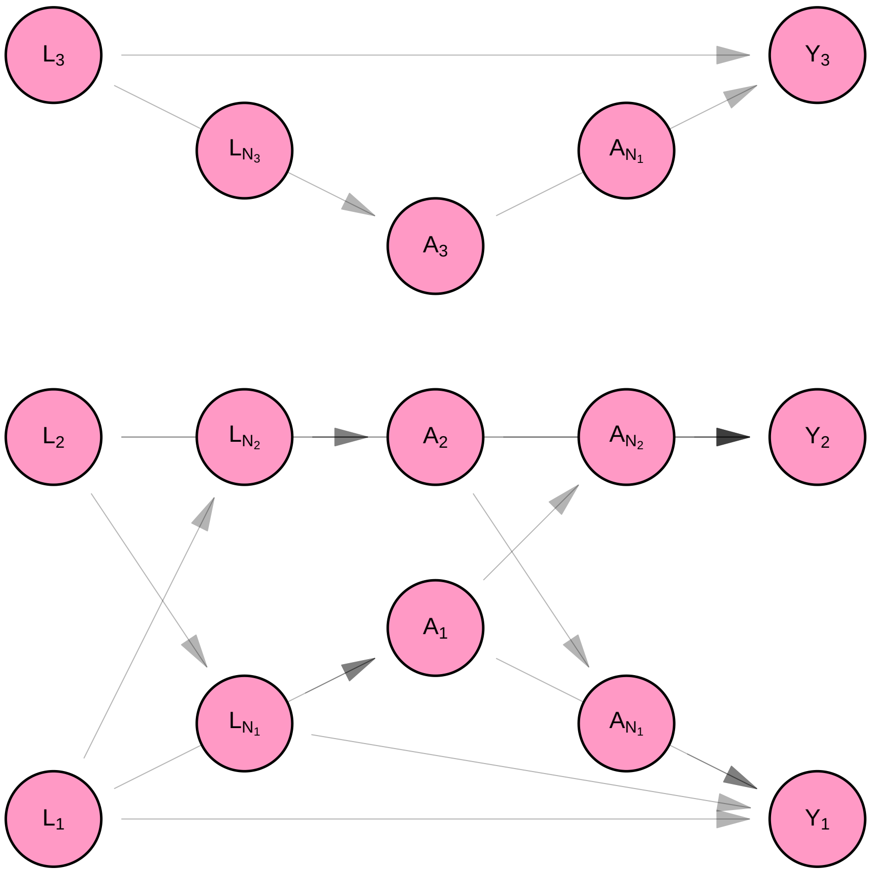

Figure E.6: Causal DAG with homogenous structure

E.2.2.2 Counterfactual data

We define counterfactual outcomes $Y_1^{(a_1,\mathbf{a}_{\mathcal{N}_1})}$, $Y_2^{(a_2,\mathbf{a}_{\mathcal{N}_2})}$, and $Y_3^{(a_3,\mathbf{a}_{\mathcal{N}_3})}$, which represent the level of job opportunities, *Persons 1*, *2*, and *3*, would have if we set their dress to level a_1, a_2, and a_3, respectively. Then the model for counterfactual variable is given by Equation E.7. It is represented using a Single World Intervention Graph (SWIG) in Figure E.7.

Note that because of the assumption of network interference, the counterfactual outcome for *Person 1* is a function of *Person 2*'s treatment value, and vice versa. The counterfactual outcome for *Person 3*, on the other hand, is a function of *Person 3*'s treatment (a_3) alone. This is because $\mathcal{N}_1 = \{2\}$, $\mathcal{N}_2 = \{1\}$, and $\mathcal{N}_3 = \{\}$, according to Example 2.

$$\begin{aligned} L_i &= f_L(U_{L_i}) \\ A_i &= f_A(U_{A_i}, L_i, g_A(\mathbf{L}_{\mathcal{N}_i})) \\ Y_i^{(a_i,\mathbf{a}_{\mathcal{N}_i})} &= f_Y(U_{Y_i}, L_i, a_i, g_Y(\mathbf{L}_{\mathcal{N}_i}), h_Y(\mathbf{a}_{\mathcal{N}_i})) \end{aligned} \tag{E.7}$$

E.2.3 Estimates

E.2.3.1 Causal estimands

We are interested in whether on average, in our population of three people, changing someone's dress would change their level of job offers To determine the answer, we will compare the average counterfactual outcome of job opportunities if everyone wore formal dress to the average counterfactual outcome of job opportunities if everyone wore non-formal dress. i.e. We will make the following comparison.

$$\frac{1}{n}\sum_i E[Y_i^{(a=1,\mathbf{a}_{\mathcal{N}_i}=\mathbf{1})} - Y_i^{(a=0,\mathbf{a}_{\mathcal{N}_i}=\mathbf{0})}]$$

It is possible to decompose this average causal effect into a portion attributable to direct (within individual) causal pathways. This tells us, on average, to what extent one's own dress affects one own's job opportunities.

$$\frac{1}{n}\sum_i E[Y_i^{(a=1,\mathbf{a}_{\mathcal{N}_i}=\mathbf{0})} - Y_i^{(a=0,\mathbf{a}_{\mathcal{N}_i}=\mathbf{0})}]$$

We can calculate the portion of the average total effect attributable to spillover (cross-individual) causal pathways. This tells us, on average, to what extent one's spouse's dress affects one's job opportunities.

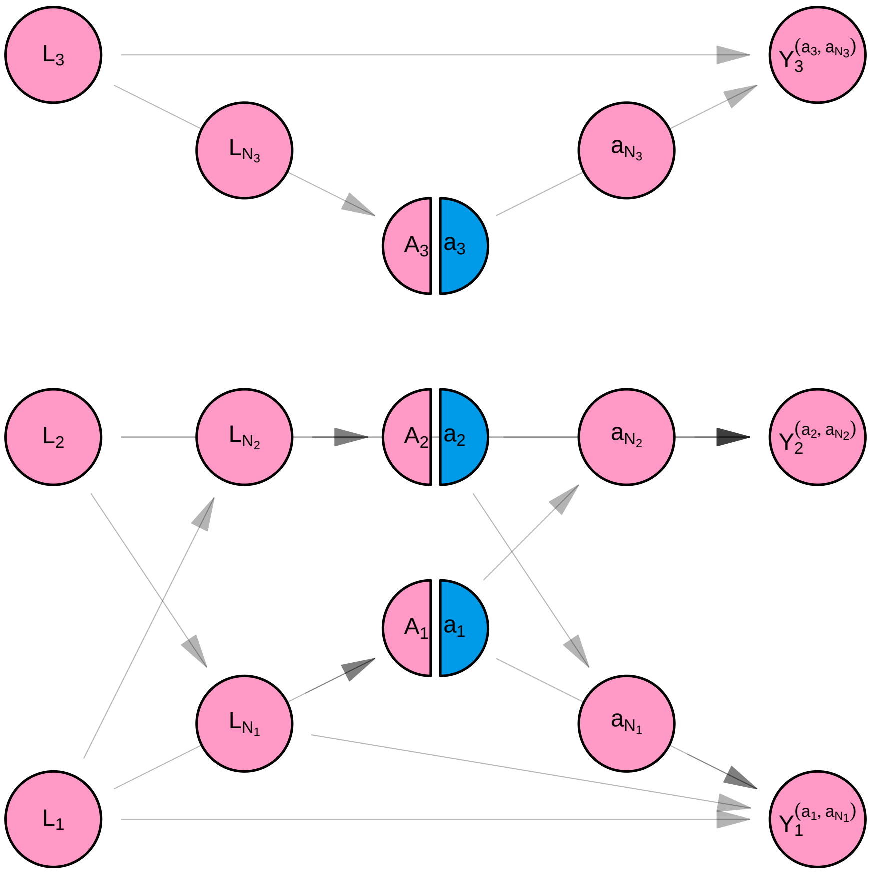

Figure E.7: SWIG with network interference

$$\frac{1}{n}\sum_i E[Y_i^{(a=1,\mathbf{a}_{\mathcal{N}_i}=\mathbf{1})} - Y_i^{(a=1,\mathbf{a}_{\mathcal{N}_i}=\mathbf{0})}]$$

E.2.3.2 Identification

$$E\left[\frac{1}{n}\sum_i Y_i^{(a_i,\mathbf{a}_{\mathcal{N}_i})}\right] = E\left[\frac{1}{n}\sum_i E[Y_i^{(a_i,\mathbf{a}_{\mathcal{N}_i})}|\mathbf{L}]\right]$$

We can start by using the law of iterated expectations. Then we can evaluate the inner expectation as follows:

$$\begin{aligned} E[Y_i^{(a_i,\mathbf{a}_{\mathcal{N}_i})}|\mathbf{L}] &= E[Y_i^{(a_i,\mathbf{a}_{\mathcal{N}_i})} \quad |L_i, g(\mathbf{L}_{\mathcal{N}_i})] \\ &= E[Y_i^{(a_i,\mathbf{a}_{\mathcal{N}_i})} \quad |L_i, g(\mathbf{L}_{\mathcal{N}_i}), A_i = a_i, h_Y(\mathbf{A}_{\mathcal{N}_i}) = h_Y(\mathbf{a}_{\mathcal{N}_i})] \\ &= E[f_Y(...) \quad |L_i, g(\mathbf{L}_{\mathcal{N}_i}), A_i = a_i, h_Y(\mathbf{A}_{\mathcal{N}_i}) = h_Y(\mathbf{a}_{\mathcal{N}_i})] \\ &= E[Y_i \quad |L_i, g(\mathbf{L}_{\mathcal{N}_i}), A_i = a_i, h_Y(\mathbf{A}_{\mathcal{N}_i}) = h_Y(\mathbf{a}_{\mathcal{N}_i})] \end{aligned}$$

- In the first line, we use the fact that $Y_i^{(a_i,\mathbf{a}_{\mathcal{N}_i})}$ is independent of L_j for $j \notin \mathcal{N}_i$ once we condition for L_i and $g(\mathbf{L}_{\mathcal{N}_i})$ - the network exposure value for person i with respect to $\mathbf{L}$.
- In the second line, we use the fact that $Y_i^{(a_i,\mathbf{a}_{\mathcal{N}_i})}$ is independent of $\mathbf{a}$ after adjusting for L, since $\mathbf{a}$ is a constant of the experimenters choosing.
- We replace $Y_i^{(a_i,\mathbf{a}_{\mathcal{N}_i})}$ with its structural equation Figure E.7, noting that this is equal to the observed value Y_i for people i whose treatment value A_i and network exposure value $h_Y(\mathbf{A}_{\mathcal{N}_i})$ is consistent with the value of $\mathbf{a}$ set by the experimenter.

We are left with this value

$$E[\frac{1}{n}\sum_i E[Y_i|L_i, g(\mathbf{L}_{\mathcal{N}_i}), A_i = a_i, h_Y(\mathbf{A}_{\mathcal{N}_i}) = h_Y(\mathbf{a}_{\mathcal{N}_i})]]$$

which we estimate as follows

$$\frac{\sum_i E[Y_i|L_i, g(\mathbf{L}_{\mathcal{N}_i}), A_i = a_i, h_Y(\mathbf{A}_{\mathcal{N}_i}) = h_Y(\mathbf{a}_{\mathcal{N}_i})] I_{\{x=a\}}(a_i) I_{\{h_Y(x)=h_Y(\mathbf{a})\}}(a_{\mathcal{N}_i})}{\sum_i I_{\{x=a\}}(a_i) I_{\{h_Y(x)=h_Y(\mathbf{a})\}}(a_{\mathcal{N}_i})}$$

Reflection questions

1. Compare and contrast the assumptions made in Section E.1 with those in Section E.2.
2. Why might it be useful for network exposure mappings to be bounded above? (hint: asymptoptics)?
3. Write an example of a network exposure mapping that is not seperable.

E.3 SSNAC

Example 3: Longitudinal Data-Driven Vaccination Campaign

You are a public health officer at the department of health in Nowhereville, and are planning to conduct and assess a three week flu vaccination campaign. None of the townspeople have been vaccinated against flu. There is a population of 5 residents in the city, and they each live in one of three neighborhoods. In addition, residents visit their friends' homes in other neighborhoods.

Person 1 lives in *Neighborhood A* and visits *Neighborhood B*. *Person 2* lives in *Neighborhood B*, but visits *Neighborhoods A* and *B*. *Person 3* lives in *Neighborhood C* and visits *Neighborhood B*. *Persons 5* and *6* live in *Neighborhood C* and they do not visit any other neighborhoods.

The city of Nowhereville has a very structured approach to vaccination campaigns. At the beginning of each week, each neighborhood gets a risk rating based on the prevalence of vaccination among its residents and visitors. Based on that rating, each neighborhood decides whether to conduct a local vaccination drive or not. Vaccination drives are done in the same way regardless of neighborhood - visitors and residents are offered immediate vaccination at a nearby mobile van through a pop-up message on the local dating app. Not everyone will receive the messages because some will not open the app during the period of the campaign. Those who do will have the choice of getting vaccinated or not. Each person can only get vaccinated once.

New tools from the SSNAC framework

As we see in Figure E.6, causal DAGs become quickly visually cluttered, and therefore unwieldy, once we start depicting causal systems with multiple observational units. If we added a fourth observational unit to Figure E.6, it would be nearly impossible to see which arrow connects which nodes This is an important limitation because one of the most useful things about causal DAGs is that they allow the observational epidemiologist to reason graphically about the structure of confounding and selection.

E.3.0.1 Introducing the Causal Directed Acyclic Network Graph (DANG)

We introduce the causal Directed Acyclic Network Graph (causal DANG) - a simple extension of the DAG that allows for the representation of complex causal systems. Like the causal DAG, the causal DANG encodes a non-parametric structural equation model.

Compact graphic representation

DANGs encode the structure of the causal process within individuals separately from the structure of the causal process across individuals. The former is termed the "structural part" of the DANG and the latter the "relational part".

We represent the structural model defined by Equation E.6 shown in Figure E.6. The first aspect (structural part) of the causal DANG shows the causal structure within individuals. It is a DAG of vector-valued variables $\mathbf{L}$, $\mathbf{A}$, and $\mathbf{Y}$ of length equal to the number of people in the sample. This DAG has standard interpretation under d-separation rules. It gives the conditional independencies across these vector-valued random variables. This is an incomplete set of independencies, however, since some of the entries in $\mathbf{A}$ are independent of a given entry in $\mathbf{L}$, and others are correlated, for instance. The second part of the causal DANG (relational part) shows which pairs of individuals have correlated measures with each other. For instance, below, we see that individuals 1 and 2 are related.

Causal DANGs give us instructions for constructing causal DAGs in the face of causal interference. For instance, according to the causal DANG shown in Figure E.8, follow these instructions to draw the DAG in Figure E.5:

- First represent all the entries of $\mathbf{L}$, $\mathbf{A}$, and $\mathbf{Y}$.
- Then for each individual $i \in \{1, 2, 3\}$, connect L_i to A_i, L_i to Y_i, and A_i to Y_i.
- Finally, for the connected pair $1 \sim 2$, connect L_1 to A_2 and L_2 to A_1; connect L_1 to Y_2, and L_2 to Y_1, and connect A_1 to Y_2 and A_2 to Y_1.

Multiple entity types

The causal DANG relaxes the assumption that all measures in the causal system can be grouped into non-overlapping sets (i.e. that each measure can be attributed to an individual in the sample). In this framework, different types of entities, represented by vectors of different lengths, can be depicted on the same DANG.

The causal DANG shown in Figure E.9 represents the first week of the campaign mentioned in Example 3. Let $\mathbf{T}$ be a 3×1 vector whose entries T_j are 1 if neighborhood j is to be vaccinated and 0 otherwise. $\mathbf{M}$ is a 5×1 vector whose entries M_i indicate whether the i^{th} person received a vaccination campaign message or not. $\mathbf{V}$ is a 5×1 vector whose entries V_i show whether participant i is vaccinated or not.

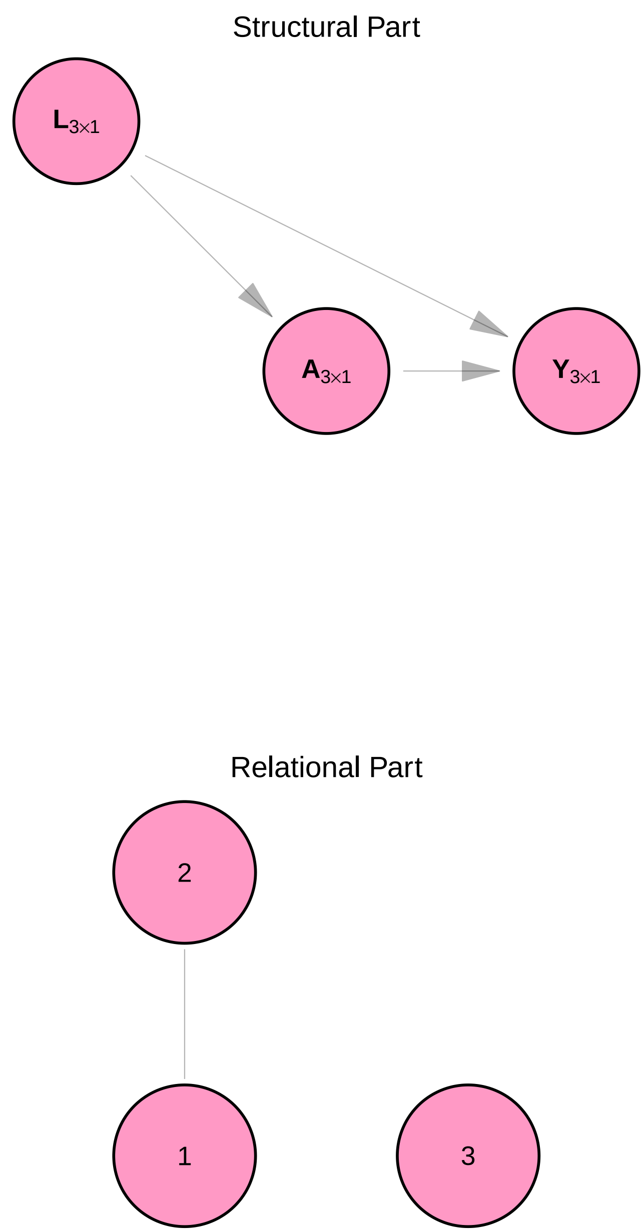

Figure E.8: Causal DANG

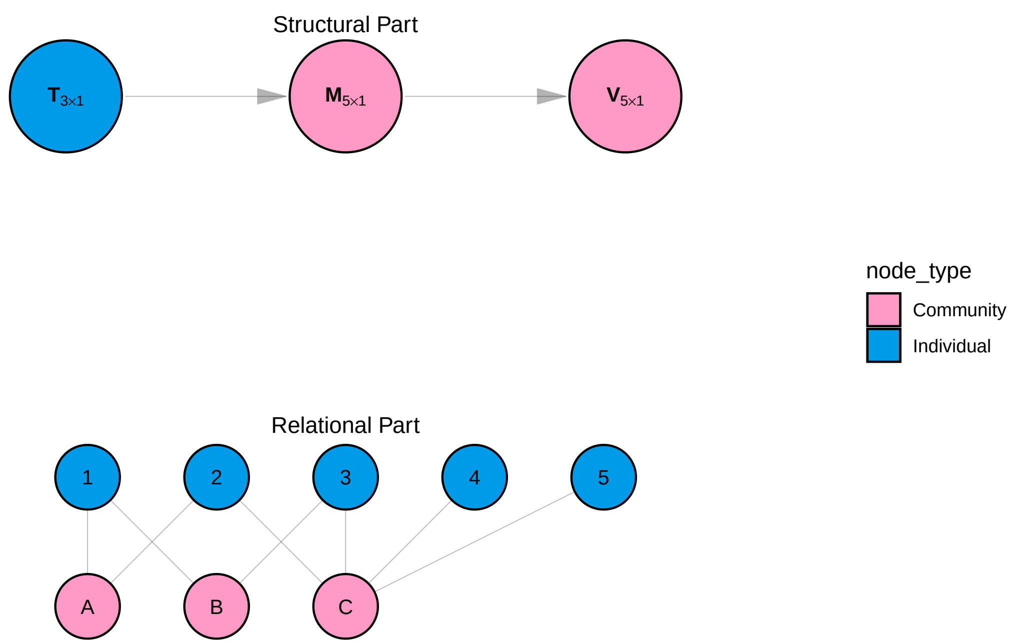

Figure E.9: Causal DANG for Week 1 in Example 3

E.3.0.2 Introducing the Single World Intervention Network Graph (SWING)

We introduce the Single World Intervention Network Graph (SWING) - a simple extension of the SWIG that allows for the representation of the counterfactual outcomes of interventions in complex causal systems.

The SWING has two parts - a structural part and a relational part. Like the causal DANG, the SWING represents vector-valued random variables (rather than scalars) as nodes. Like the SWIG, the SWING splits the nodes that are to be intervened upon by the experimenter. Figure E.10 shows the counterfactual distribution of the variables in the model under an intervention setting $\mathbf{T}$ to the value $\mathbf{t}$.

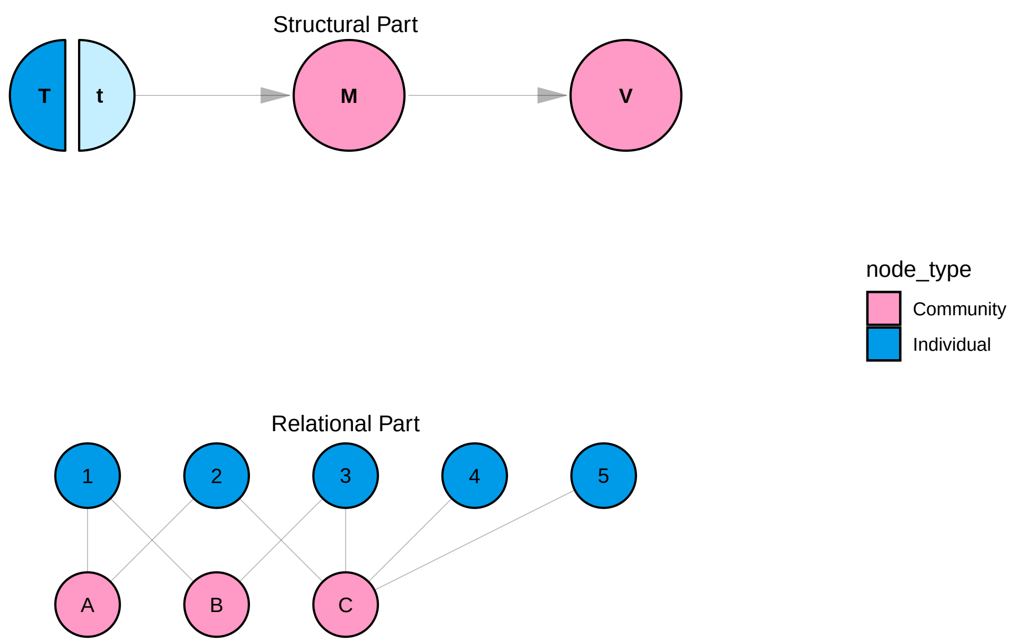

Figure E.10: SWING for Week 1 in Example 3

E.3.1 Assumptions

See Section E.2 for a fuller description of the below assumptions.

E.3.1.1 Independence assumptions

Under the SSNAC framework, we make the assumption of *no spillover*: contemporaneous measurements on observational units of a given entity type are independent of each other given all prior measures. The time-ordering of the measurements is given by the structural part of the causal DANG. We further make the assumption of *network interference* - that interference respects the mapping given by the relational part of a causal DANG.

E.3.1.2 Homogeneity assumptions

We make the *Identical distribution* assumption: the structural equations that produce each observational unit's measures are identical across observational units of the same entity type. We assume that the exposure mapping for each variable in the model is characterized by *homogeneous interference*. i.e: the exposure mapping is separable, equivalent, symmetric, and bounded. In applications with repeated measures, if the structure of the causal DANG allows, the SSNAC framework allows for the assumption of *time homogeneity*: the structural equations are identical across time. We apply this assumption in Example 3 below.

E.3.2 Model

E.3.2.1 Observational data

The causal DANG shown in Figure E.11 represents the three week campaign mentioned in Example 3 (we omit the relational part and show only the structural part). Let $\mathbf{T}_s$ be a 3×1 vector whose entries T_{sj} are 1 if *Neighborhood* j is to be vaccinated at time s and 0 otherwise. $\mathbf{M}_s$ is a 5×1 vector whose entries M_{si} indicate whether the i^{th} person received a campaign message at time s or not. $\mathbf{V}_s$ is a 5×1 vector whose entries V_{si} show whether participant i is vaccinated or not at time s. Finally, $\mathbf{R}_s$ is a 3×1 vector whose entries R_{si} shows the risk rating of neighborhood j at the beginning of time s. Equation E.8 specifies this model for *Person* i and *Neighborhood* j at time s:

$$\begin{aligned}
M_{0i} &= f_{M_0}(U_{M_{0i}}) \\
V_{0i} &= f_{V_0}(U_{V_{0i}}) \\
R_{sj} &= f_R(U_{R_{sj}}, \mathbf{V}_{s-1}) \\
T_{sj} &= f_T(U_{T_{sj}}, R_{s,j}) \\
M_{si} &= f_M(U_{M_{\{si\}}}, V_{\{s-1,i\}}, M_{\{s-1,i\}}, \mathbf{T}_{\{s,\mathcal{N}_i\}}) \\
V_{si} &= f_V(U_{V_{\{si\}}}, M_{\{si\}}, V_{\{s-1,i\}})
\end{aligned} \tag{E.8}$$

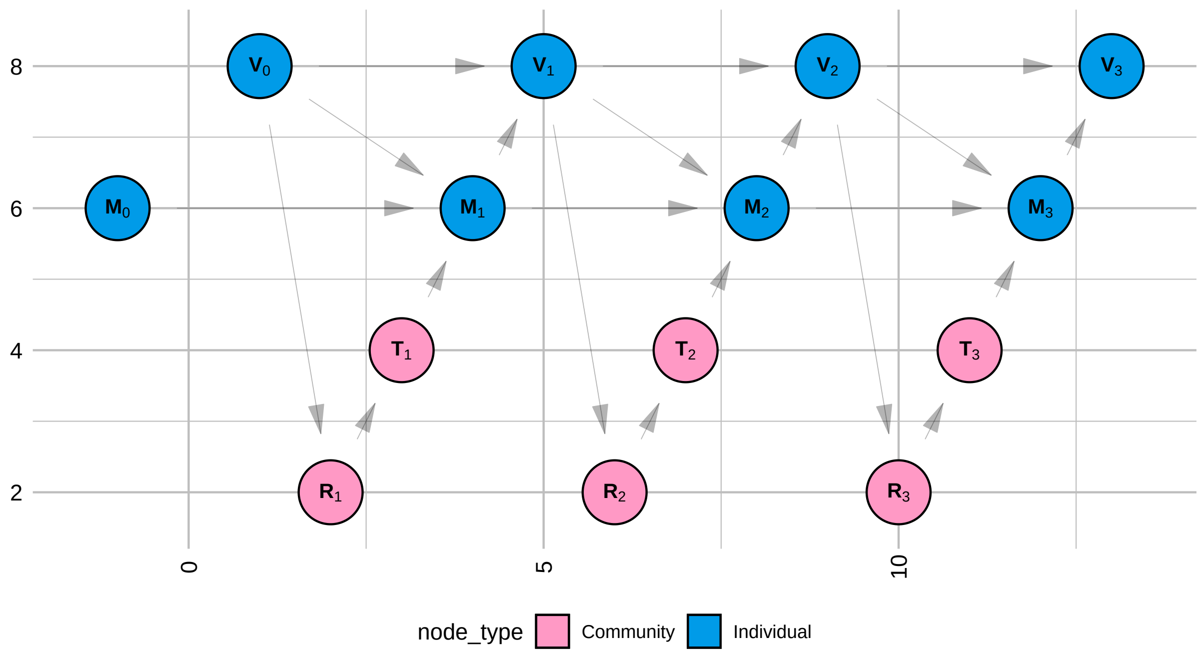

Figure E.11: Causal DANG for Week 1-3 in Example 3

According to Equation E.8, R_{sj}, the risk rating for *Neighborhood* j at time s, is a function of $\mathbf{V}_{s-1}$, the vaccination status of every individual at time $s-1$. T_{sj} - an indicator for whether or not *Neighborhood* j conducted a campaign at time s, is a function of the risk rating for *Neighborhood* j at time s.

M_{si}, which indicates whether or not an *Person* i received a campaign message at time s, is a function of whether or not person i was vaccinated at time $s-1$, whether they received a campaign message at time $s-1$, and whether or not one of the neighborhoods they live in or visit conducted a campaign at time s. The vaccination status of individual i at time s is a function of whether or not they received a campaign message at time s and whether or not they were vaccinated at time $s-1$.

E.3.2.2 Counterfactual data

Say we want to understand the impact of a vaccination campaign conducted over three weeks on vaccination status. This would be done by setting $\mathbf{T}_1$ to $\mathbf{t}_1$, $\mathbf{T}_2$ to $\mathbf{t}_2$, and $\mathbf{T}_3$ to $\mathbf{t}_3$ and noting the resulting counterfactual vaccination status at the end of week 1 ($\mathbf{V}_1^{(\mathbf{t}_1)}$), week 2 ($\mathbf{V}_1^{(\mathbf{t}_1,\mathbf{t}_2)}$), and week 3 ($\mathbf{V}_1^{(\mathbf{t}_1,\mathbf{t}_2,\mathbf{t}_3)}$). This counterfactual model is defined Equation E.9 as shown in shown in Figure E.12.

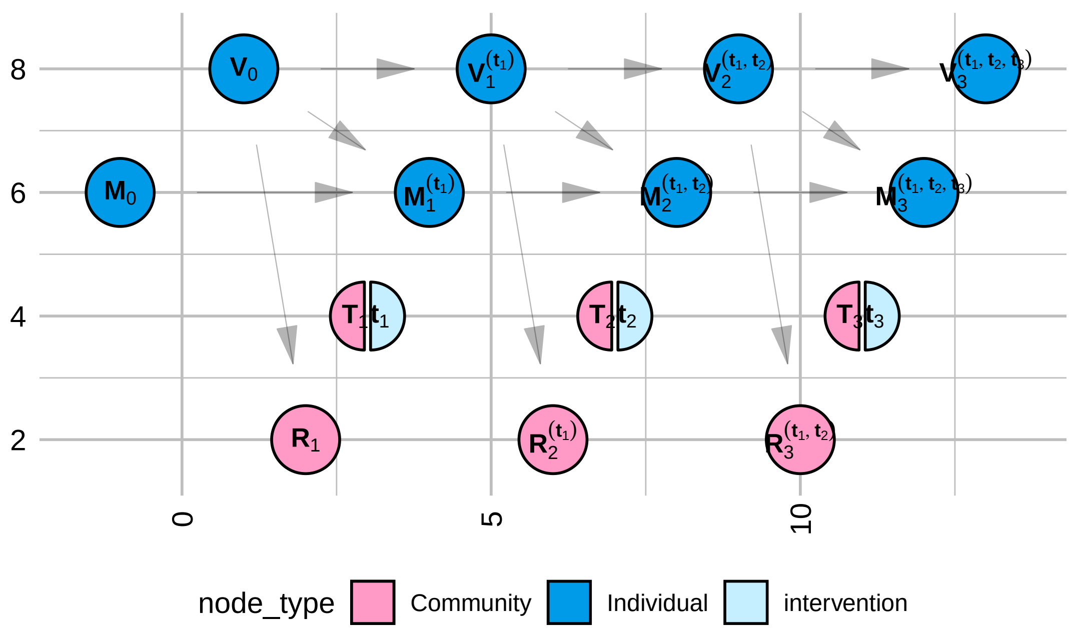

Figure E.12: SWING for Week 1-3 Example 3

$$
\begin{aligned}
M_{0i} &= f_{M_0}(U_{M_{0i}}) \\
V_{0i} &= f_{V_0}(U_{V_{0i}}) \\
R_{sj}^{(\bar{\mathbf{t}})} &= f_R(U_{R_{sj}}, \mathbf{V}_{s-1}^{(\bar{\mathbf{t}})}) \\
T_{sj}^{(\bar{\mathbf{t}})} &= f_T(U_{T_{sj}}, R_{s,j}^{(\bar{\mathbf{t}})}) \\
M_{si}^{(\bar{\mathbf{t}})} &= f_M(U_{M_{\{si\}}}, V_{\{s-1,i\}}^{(\bar{\mathbf{t}})}, M_{\{s-1,i\}}^{(\bar{\mathbf{t}})}, \mathbf{t}_{\{s,\mathcal{N}_i\}}) \\
V_{si}^{(\bar{\mathbf{t}})} &= f_V(U_{V_{\{si\}}}, M_{\{si\}}^{(\bar{\mathbf{t}})}, V_{\{s-1,i\}}^{(\bar{\mathbf{t}})})
\end{aligned}
\tag{E.9}
$$

where $\bar{\mathbf{t}}$ is defined as "the history of $\mathbf{t}$". i.e. at time 1, $\bar{\mathbf{t}} = \mathbf{t}_1$. At time 2, $\bar{\mathbf{t}} = [\mathbf{t}_1, \mathbf{t}_2]$, etc.

E.3.3 Estimates

E.3.3.1 Causal estimands

We are interested in vaccine coverage throughout the three week campaign. For a given week s, this is the sum of the vaccine status outcome $\mathbf{V}$ across individuals.

$$
E[\sum_{i=1}^{5} V_{si}^{(\bar{\mathbf{t}})}]
$$

E.3.3.2 Identification

Because individual vaccination status is both a confounder and a mediator of the relationship between neighborhood intervention status and vaccination status, we cannot identify our estimands using conditional expectations of the observed data. We must use a marginal structural model. See Aronow and Samii (2017).

We can estimate the outcome of interest using the inverse probability of treatment weights, noting that the treatment in this case is the network exposure value $g(t_{\bar{s},N_i})$ for $\mathbf{T}$.

$$
V_{si}^{(\bar{\mathbf{t}})} = \frac{V_{si}\mathbf{1}_{x=g(t_{\bar{s},N_i})}(g(T_{\bar{s},N_i}))}{\prod_{q=1}^{s} P[g(\mathbf{T}_{q,N_i}) = g(\mathbf{t}_{q,N_i})|\mathbf{R}_{\bar{q}N_i}]}
$$

where $\mathbf{R}_{\bar{s}N_i}$ is the "history of $\mathbf{R}_{sN_i}$". e.g.

$$
\mathbf{R}_{\bar{3}N_i} = [\mathbf{R}_{1N_i}, \mathbf{R}_{2N_i}, \mathbf{R}_{3N_i}]
$$

Reflection questions

1. Compare and contrast the assumptions made in Section E.2 with those made in Section E.3.
2. What is the difference between a causal DAG and a causal DANG?
3. What is the difference between a SWIG and a SWING?

E.4 Outbreak control intervention

We used data from the MPX NYC study to assess the effectiveness of two approaches to conducting mass vaccination campaigns quickly. MPX NYC is an online survey of about 1300 individuals that was conducted in 2022 in New York City. Study participants were asked to indicate their home on a map, and were asked to indicate places they had had social contact in a congregate setting. Spatial coordinates were converted into community district identifiers. Community districts spatially partition the city of New York.

Application: Outbreak control model using MPX NYC data

Under the SSNAC framework, our intervention of interest might not be about setting the value of any particular variable, but setting a policy or a process that produces a set of variables in the causal model. This would be done by changing the structural equations which produce those variables. In the SWING shown in Figure E.13 and Equation E.10, we show an intervention which replaces the function for the risk rating for each community ($\mathbf{R}$) with the function h. Every variable that is downstream of the first variable created using h is marked with a superscript (h) to show that it is partially determined by that function.

We can now ask what the impact of taking policy h would be on the decision whether or not to vaccinate a neighborhood j at time s (T_{js}). We can also ask what the impact of the policy is on the the exposure of participant i to the campaign message at time s (M_{is}).

Our interest in the MPX NYC project, however, was initially to provide evidence to inform a targeted vaccination campaign, so the main outcomes of interest are the vaccination status of each individual i at time s ($V_{is}^{(h)}$). We were interested in contrasting two approaches to assigning risk-ratings to community district: the contact-neutralizing approach, and the movement-neutralizing approach.

Whereas the contact-neutralizing approach ranks community districts by how many individuals are connected to them, the movement-neutralizing approach ranks community districts by how centrally they are connected to other community districts through the mutual connection with certain individuals.

E.4.1 Assumptions

We make the simplifying assumption that only one neighborhood is vaccinated at a time - the top-ranked one. We make the further simplifying assumptions that every person associated with a community district (either through residence or social activity) receives the campaign message when a community district is vaccinated, and that every person who receives a campaign message gets vaccinated. Once an individual is vaccinated, they remain vaccinated.

E.4.2 Model

According to the model specified by structural equations E.10, no individual i has been vaccinated at time 0, and nobody receives the campaign message. The risk rating for community district j at time s is some function h of the vaccination status at time $s-1$ for each individual who is connected to that community district. An intervention is conducted in community district j as time s if that community district has the highest risk rating. Each individual i is said to have received the campaign message at time s if the intervention was conducted in any of the community districts he is connected to. Finally, we consider individual i vaccinated at time s if they were vaccinated at time $s-1$ or if they received the campaign message in time s. The risk rating is given by the function h.

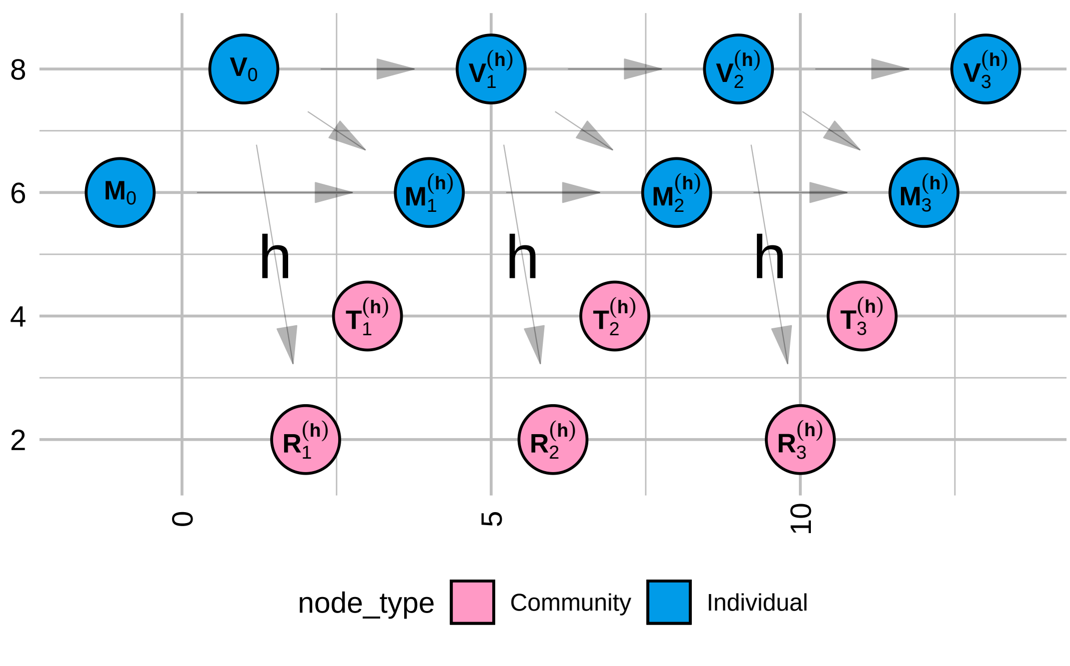

Figure E.13: SWING for Week 1-3 for MPX NYC Application of SSNAC framework

$$
\begin{aligned}
M_{0i} &= 0 \\
V_{0i} &= 0 \\
R_{sj}^{(h)} &= h(\mathbf{V}_{s-1}) \\
T_{sj}^{(h)} &= \begin{cases} 1 & R_{sj}^{(h)} = max_k R_{sk}^{(h)} \\ 0 & \text{otherwise} \end{cases} \\
M_{si}^{(h)} &= \begin{cases} 1 & \mathbf{T}_{s\mathcal{N}_i}^{(h)} \neq \mathbf{0} \\ 0 & \text{otherwise} \end{cases} \\
V_{si}^{(h)} &= \begin{cases} 1 & M_{si}^{(h)} = 1 \text{ or } V_{s-1,i}^{(h)} = 1 \\ 0 & \text{otherwise} \end{cases}
\end{aligned}
\tag{E.10}
$$

Define $\psi(i, j)$ as an adjacency function - an indicator function that equals to 1 if Person i lives in or visits neighborhood j and 0 otherwise. Let $\mathcal{N}_P$ be the indices of all person nodes and $\mathcal{N}_N$ be a set of indices for neighborhood nodes. Under the contact-neutralizing approach

$$
h_{sj}(\mathbf{V}_{s-1}) = \sum_{i \in \mathcal{N}_P} (1 - V_{(s-1),i}) \times \psi(i, j)
$$

and under the movement-neutralizing approach

$$
h_{sj}(\mathbf{V}_{s-1}) = \sum_{i \in \mathcal{N}_P} (1 - V_{(s-1),i}) \times \psi(i, j) \sum_{k \in \mathcal{N}_N} \psi(i, k)
$$

E.4.3 Estimands

E.4.3.0.1 Spatial efficiency

Intervention coverage for the s^{th} community district to be immunized, is defined

$$
C_s^{(h)} = \frac{1}{|\mathcal{N}_p|} \sum_i V_{si}^{(h)}
$$

The cumulative intervention coverage after immunizing s community districts is therefore

$$
\bar{C_s}^{(h)} = \sum_{v=1}^{s} C_v{}^{(h)}
$$

E.4.3.0.2 Social network impact

Define graph $G^s(N, \psi_s)$ such that

$$\psi_s(i,j) = \psi(i,j) \times (1 - V_{(s-1),i})$$

The cumulative social network impact of immunizing s community districts is measured in terms of the size of the largest of the connected components $LCC_{(s)}$ of the $G^s(N, \psi_s)$ network as a proportion of the total number of participants.

$$\frac{LCC_s}{|N_p|}$$

E.4.4 Estimation

We simulated the intervention outlined above using MPX NYC data by calculating the $G^s(N, \psi_s)$ where s ranges from 1 to the total number of community districts. We measured the uncertainty of our estimates by re-sampling study participants with replacement.